

The Pol-InSAR Course

Part 8: Pol-InSAR 1

Prepared by DLR-HR's Pol-InSAR Team

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ESAMAAP

EEBIOMASS

SAR Polarimetry (PolSAR)

Allows the identification / decomposition of different scattering processes occurring inside the resolution cell



SAR Interferometry (InSAR)

Allows the location of the effective scattering center inside the resolution cell

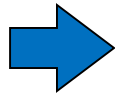
Polarimetric SAR Interferometry (Pol-InSAR)

Potential to separate in height different scattering processes occurring inside the resolution cell.

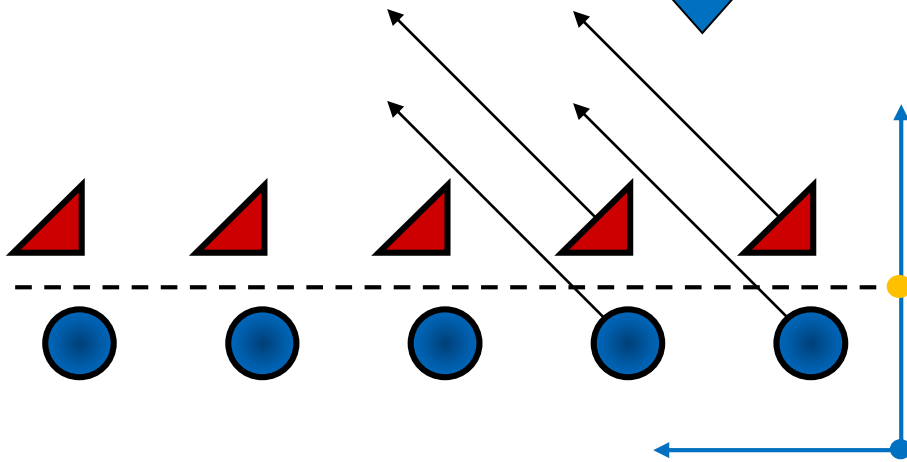
(Scalar) Interferometry vs. Polarimetry

$$S_{HH}^1 = A_D^1 + A_S^1$$

$$S_{HH}^2 = A_D^2 + A_S^2$$



$$\varphi = \arg\{ S_{HH}^1 S_{HH}^{2*} \}$$



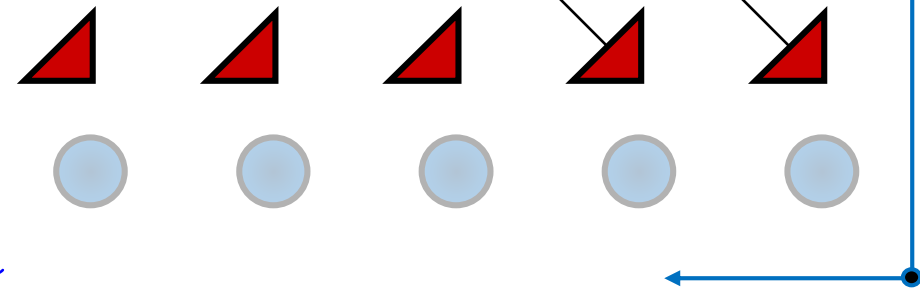
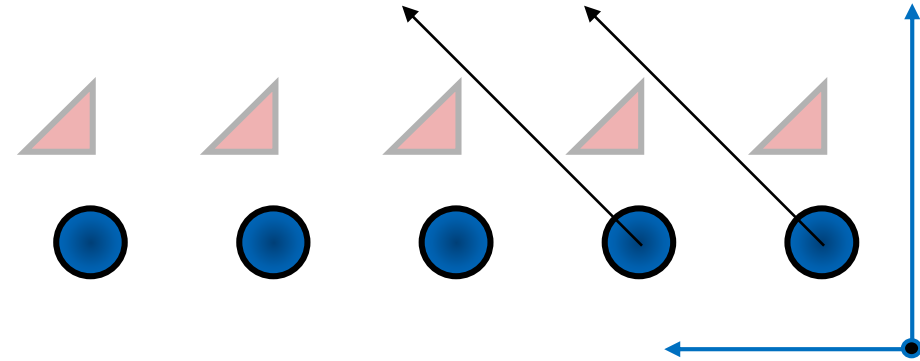
Scene

$$[S_D] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = A_D \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{Shifted in phase}$$

$$[S_S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = A_S \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{In-phase}$$

$$S_{HH} + S_{VV} = 2A_S$$

$$S_{HH} - S_{VV} = 2A_D$$



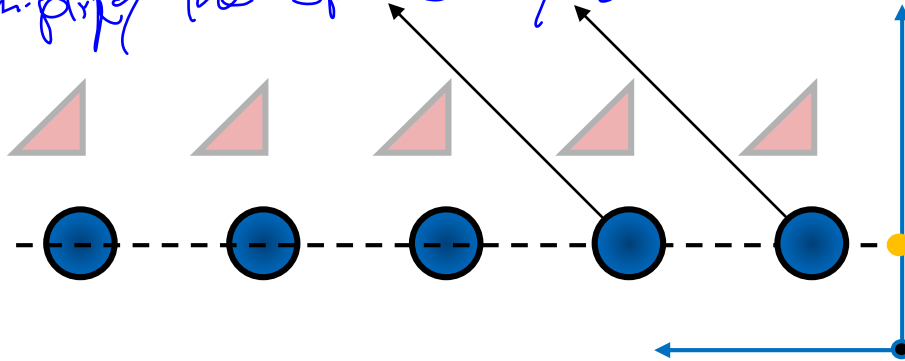
D: Dihydals
S: Spheres

Polarimetric Interferometry

$$i_{1S} = S_{HH}^1 + S_{VV}^1 = 2A_S^1$$

$$i_{2S} = S_{HH}^2 + S_{VV}^2 = 2A_S^2$$

Use path to
coupling the spheres layer



Do interferometry

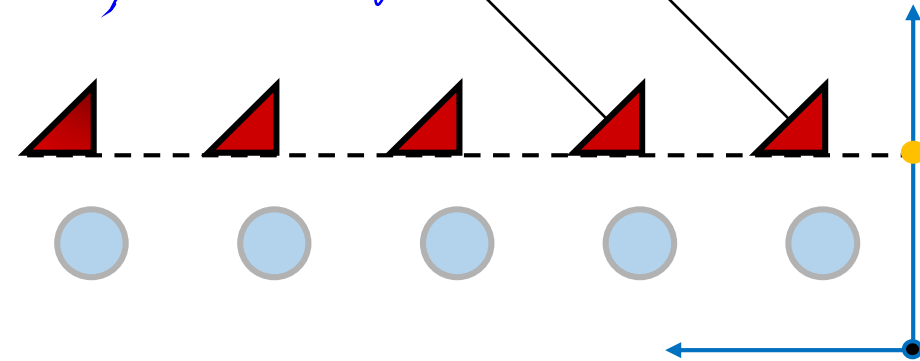
$$\varphi_S = \arg\{ i_{1S} i_{2S}^* \}$$

$$i_{1D} = S_{HH}^1 - S_{VV}^1 = 2A_D^1$$

$$i_{2D} = S_{HH}^2 - S_{VV}^2 = 2A_D^2$$

Do differential

$$\varphi_D = \arg\{ i_{1D} i_{2D}^* \}$$



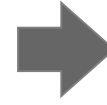
$$\begin{bmatrix} \triangle \end{bmatrix} [S_D] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = A_D \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \circ \end{bmatrix} [S_S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} = A_S \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$[S] = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}$$

*we can take it as interpretation
coherence taken at the same
time*



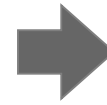
Polarimetric Coherences

$$\tilde{\gamma}(S_{ij} S_{mn}) = \frac{\langle S_{ij} S_{mn}^* \rangle}{\sqrt{\langle S_{ij} S_{ij}^* \rangle \langle S_{mn} S_{mn}^* \rangle}}$$

PolSAR



$$[S_1 \ S_2]$$



Interferometric Coherences

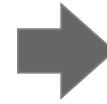
$$\tilde{\gamma}(S_1 S_2) = \frac{\langle S_1 S_2^* \rangle}{\sqrt{\langle S_1 S_1^* \rangle \langle S_2 S_2^* \rangle}}$$

InSAR



$$[S_1] = \begin{bmatrix} S_{HH}^1 & S_{HV}^1 \\ S_{VH}^1 & S_{VV}^1 \end{bmatrix}$$

$$[S_2] = \begin{bmatrix} S_{HH}^2 & S_{HV}^2 \\ S_{VH}^2 & S_{VV}^2 \end{bmatrix}$$



Polarimetric / Interferometric Coherences

$$\tilde{\gamma}(S_{ij}^1 S_{mn}^2) = \frac{\langle S_{ij}^1 S_{mn}^{2*} \rangle}{\sqrt{\langle S_{ij}^1 S_{ij}^{1*} \rangle \langle S_{mn}^2 S_{mn}^{2*} \rangle}}$$

Pol-InSAR

*with baseline
two times*

Complex Coherences on the Unit Circle (UC)

$$\tilde{\gamma} := \frac{\sum_{k=1}^N S_1(k) S_2^*(k)}{\sqrt{\sum_{k=1}^N S_1(k) S_1^*(k) \sum_{k=1}^N S_2(k) S_2^*(k)}} = \exp(i\varphi_{\text{Int}}) \cdot |\tilde{\gamma}| = \exp(i\varphi_{\text{Int}}) \cdot \gamma$$

Correlation Coefficient

$$0 \leq \gamma \leq 1$$

Interferometric Phase

$$0 \leq \varphi_{\text{Int}} \leq 2\pi$$

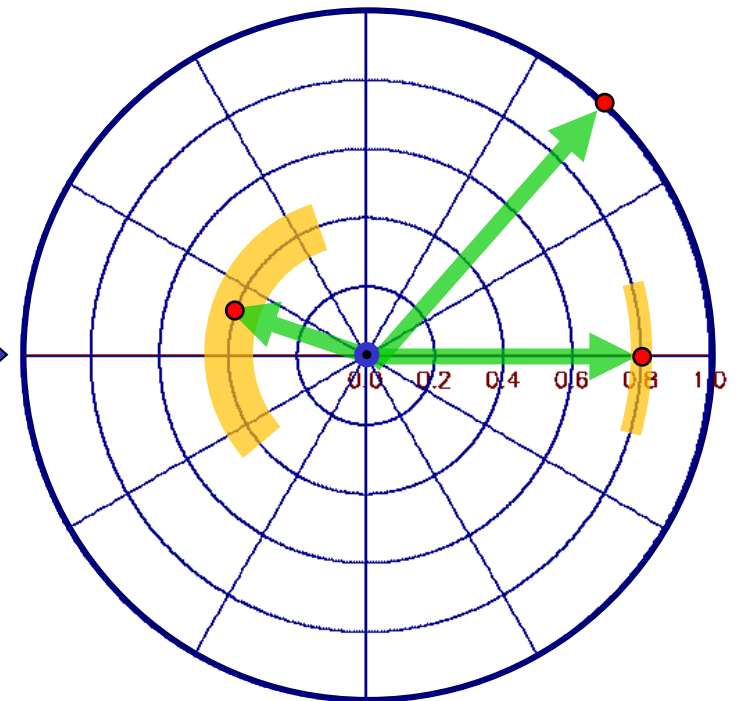
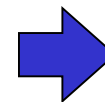
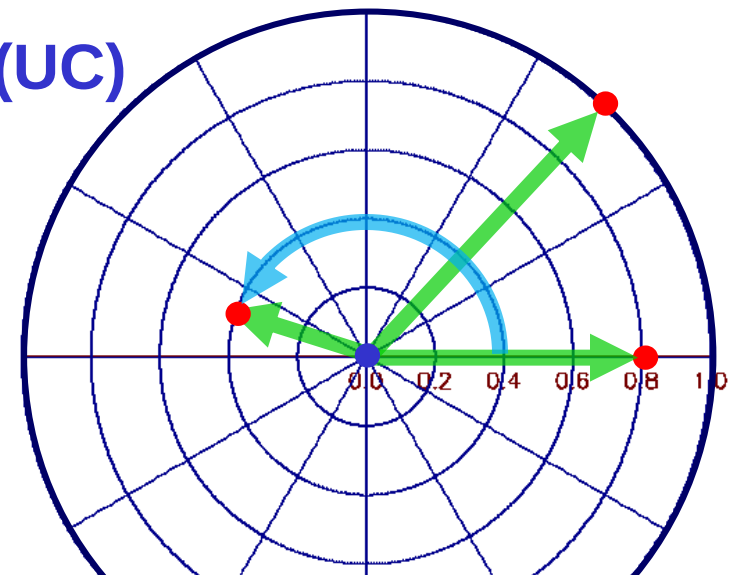
Cramer Rao Bounds:

*Coherence is now estimated
: a stochastic value*

Correlation Coefficient $\text{VAR}(\gamma)_{\text{CR}} = \frac{(1-\gamma^2)^2}{2N}$

Interferometric Phase $\text{VAR}(\varphi_{\text{Int}})_{\text{CR}} = \frac{1-\gamma^2}{2N\gamma^2}$

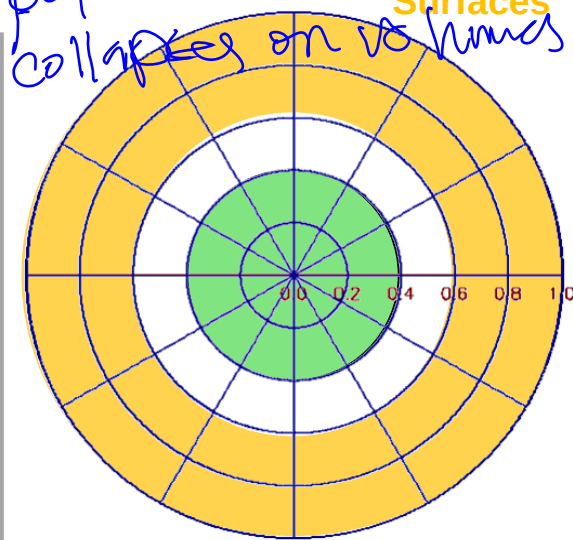
$\varphi_{\text{Int}} = \arg(\tilde{\gamma})$ and N is the number of Looks



Why is Interferometry important for Volume Scatterers?

Volumes
Surfaces

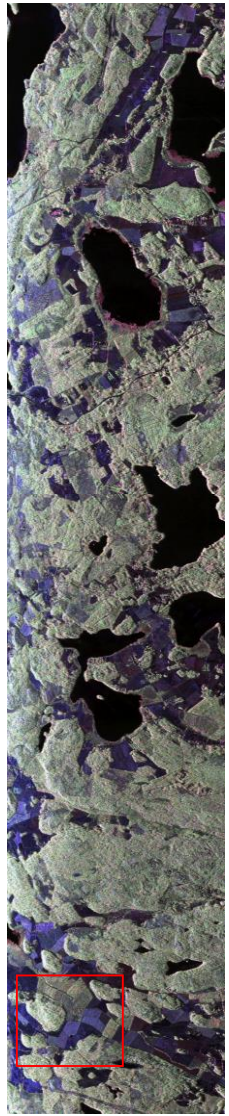
pol-coherence collapses on volumes



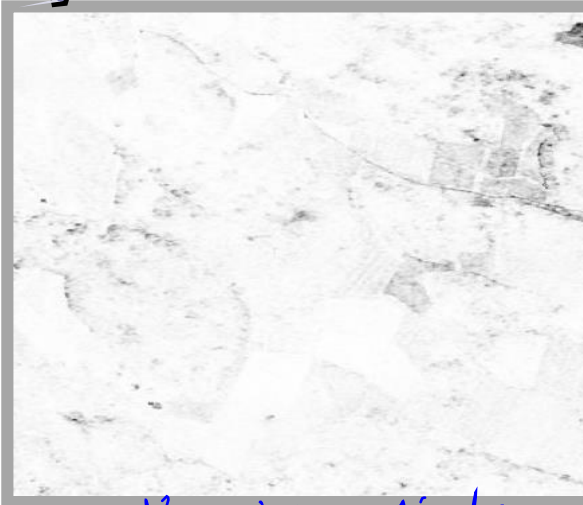
HH-VV Coherence



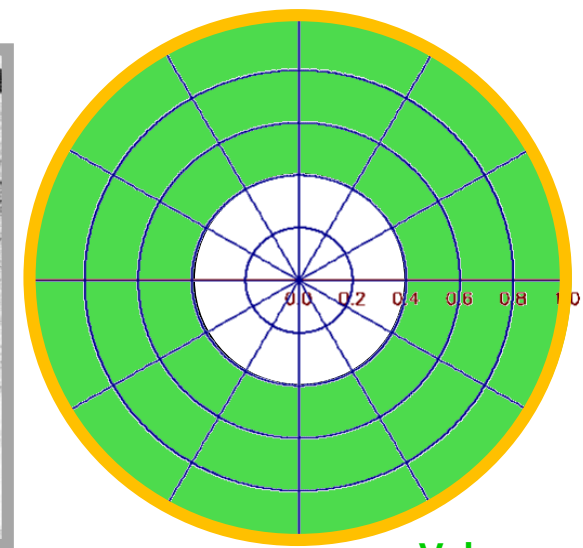
E-SAR / Test Site: Helsinki, Finland



HH-HH Coherence

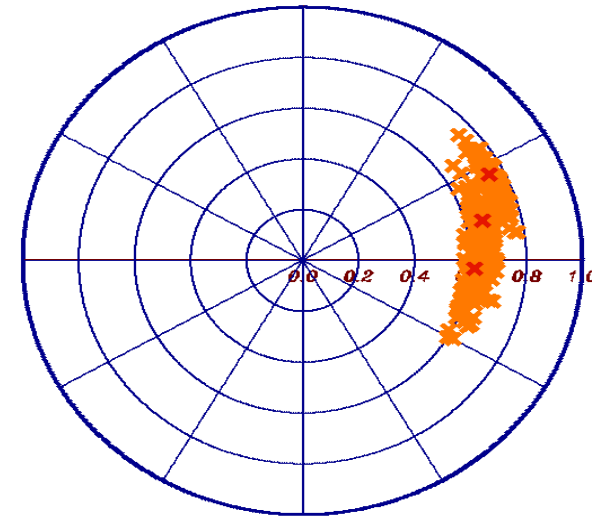
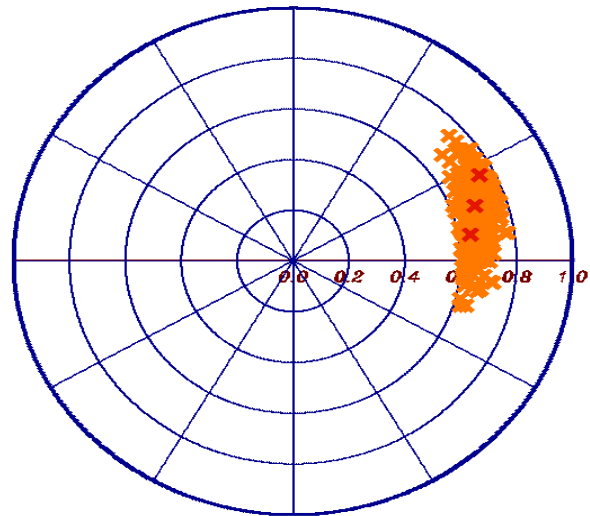
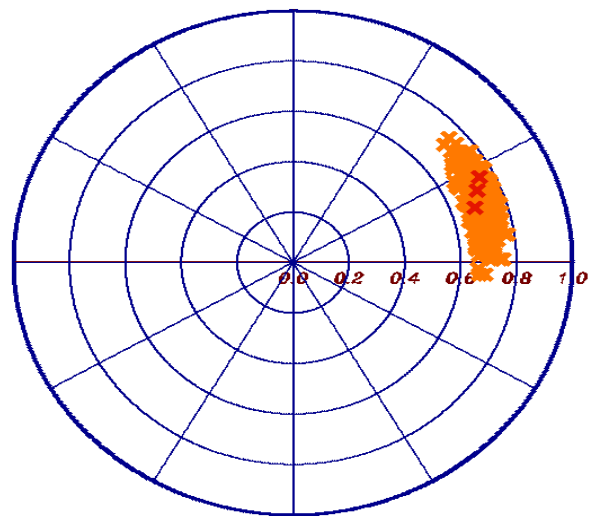


HH-HH Coherence

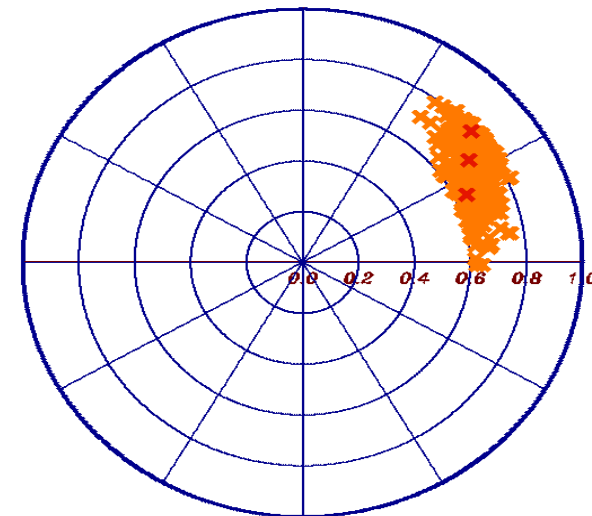
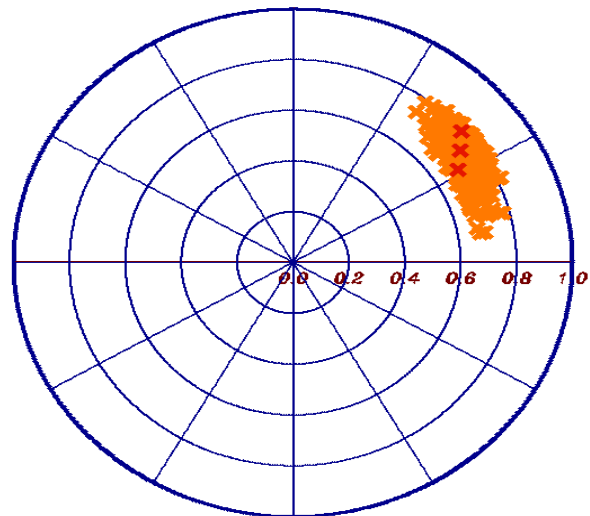
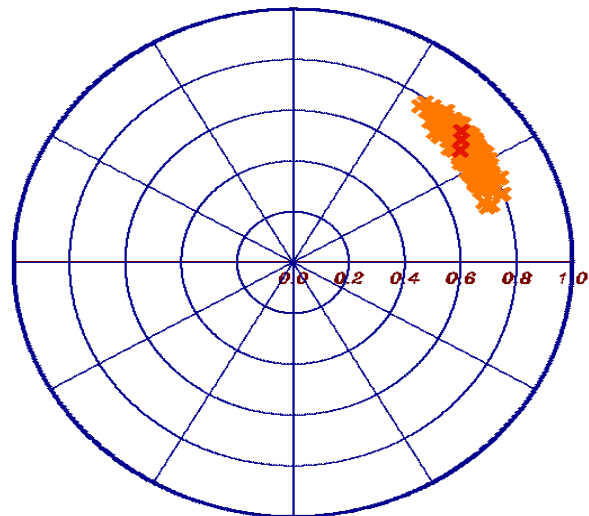


Volumes
Surfaces

*reliable phase estimate
allows us bring back
the coherence*



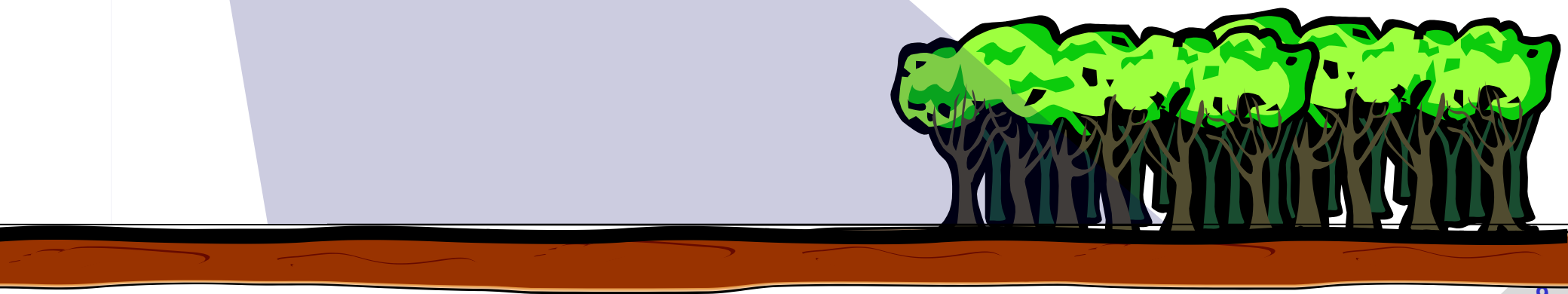
Pol-InSAR: Basic Principles & Ideas





S_1

SAR Interferometry for Volume Structure





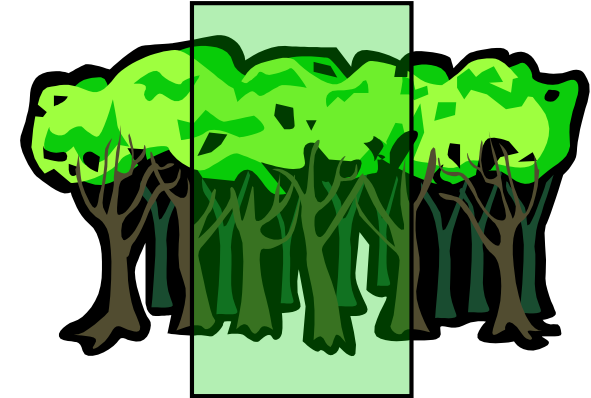
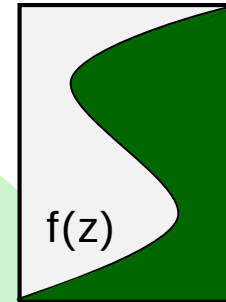
Interferometric
Coherence

$$\tilde{\gamma}(S_1, S_2) = \frac{\langle S_1 S_2^* \rangle}{\sqrt{\langle S_1 S_1^* \rangle \langle S_2 S_2^* \rangle}}$$

SAR Interferometry for Volume Structure

Volume
Coherence

$$\tilde{\gamma}_{\text{Vol}}(f(z), k_z) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z) e^{ik_z z} dz}{\int_0^{h_v} f(z) dz}$$



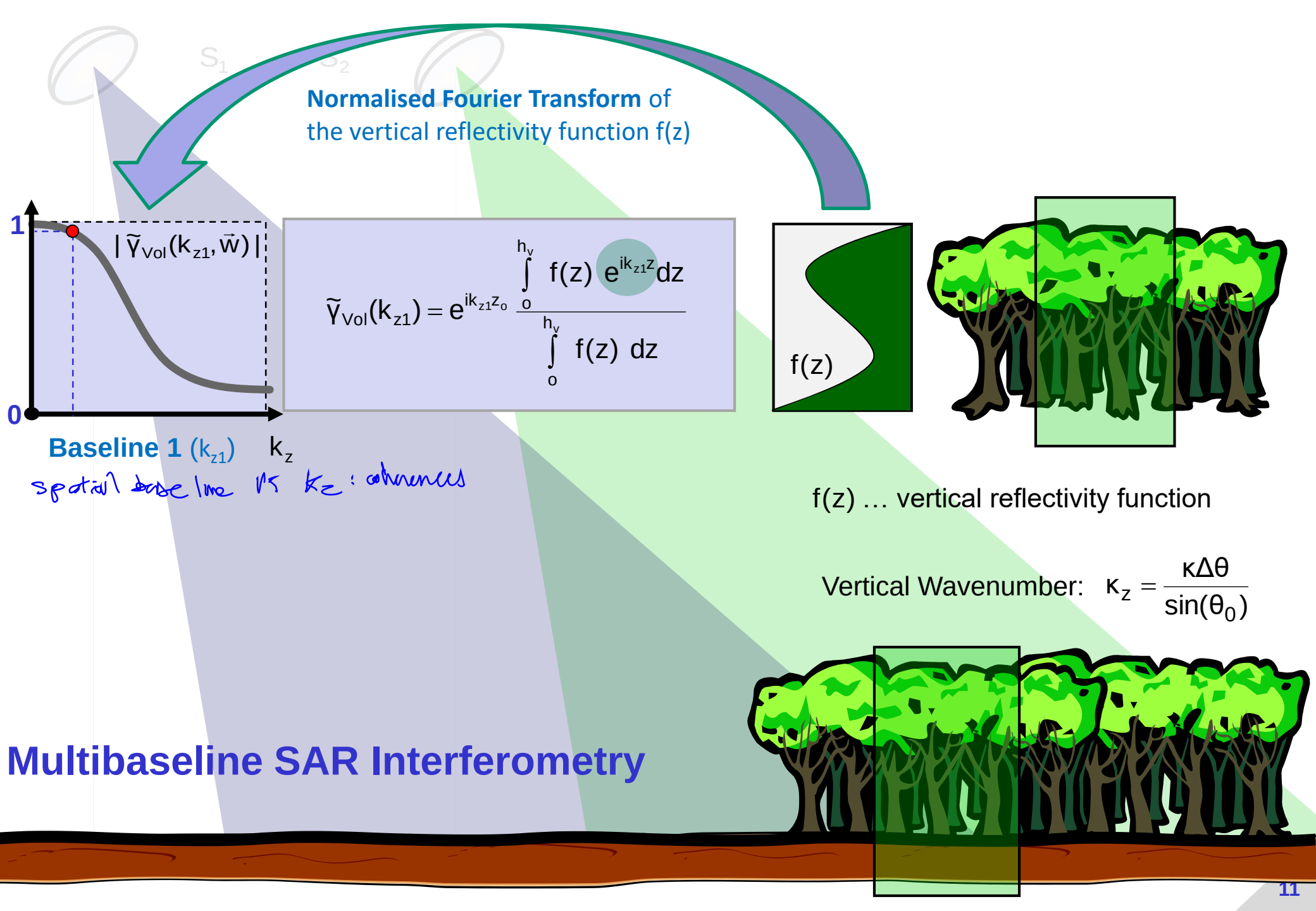
$f(z)$... vertical reflectivity function

Vertical Wavenumber: $k_z = \frac{\kappa \Delta \theta}{\sin(\theta_0)}$

$$\tilde{\gamma} = \tilde{\gamma}_{\text{Temporal}} \gamma_{\text{SNR}} \tilde{\gamma}_{\text{Vol}}$$

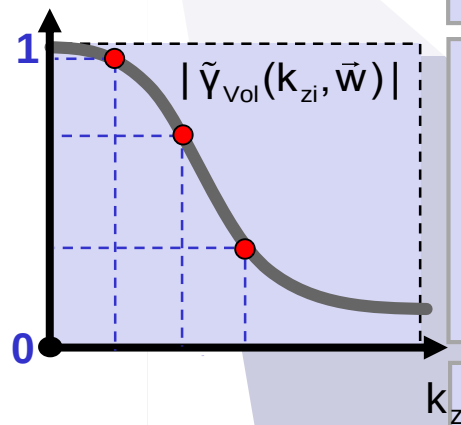
- $\tilde{\gamma}_{\text{Temporal}}$... temporal decorrelation
- γ_{SNR} ... additive noise decorrelation
- $\tilde{\gamma}_{\text{Volume}}$... geometric decorrelation







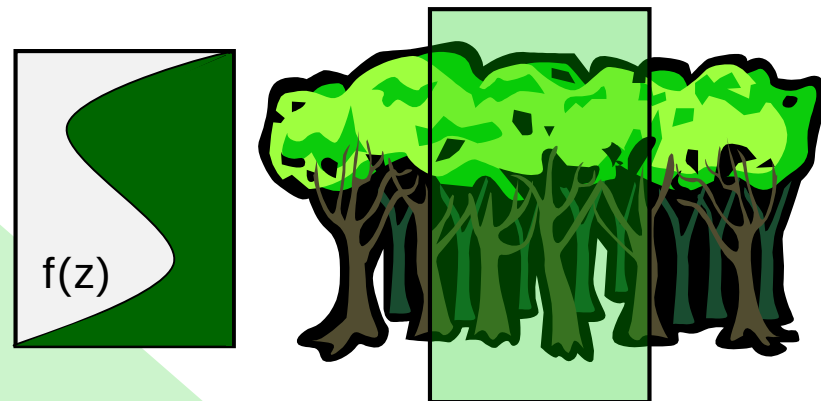
$$\tilde{Y}_{Vol}(k_{z3}) = e^{ik_{z3}z_0} \frac{\int_0^{h_y} f(z) e^{ik_{z3}z} dz}{\int_0^{h_y} f(z) dz}$$



Baseline 2 (k_{z2})

$$\tilde{Y}_{Vol}(k_{z1}) = e^{ik_{z1}z_0} \frac{\int_0^{h_y} f(z) e^{ik_{z1}z} dz}{\int_0^{h_y} f(z) dz}$$

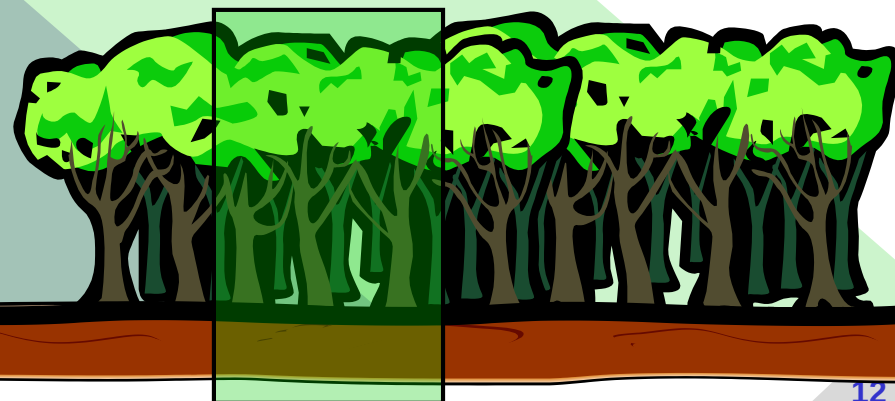
$$\tilde{Y}_{Vol}(k_{z2}) = e^{ik_{z2}z_0} \frac{\int_0^{h_y} f(z) e^{ik_{z2}z} dz}{\int_0^{h_y} f(z) dz}$$



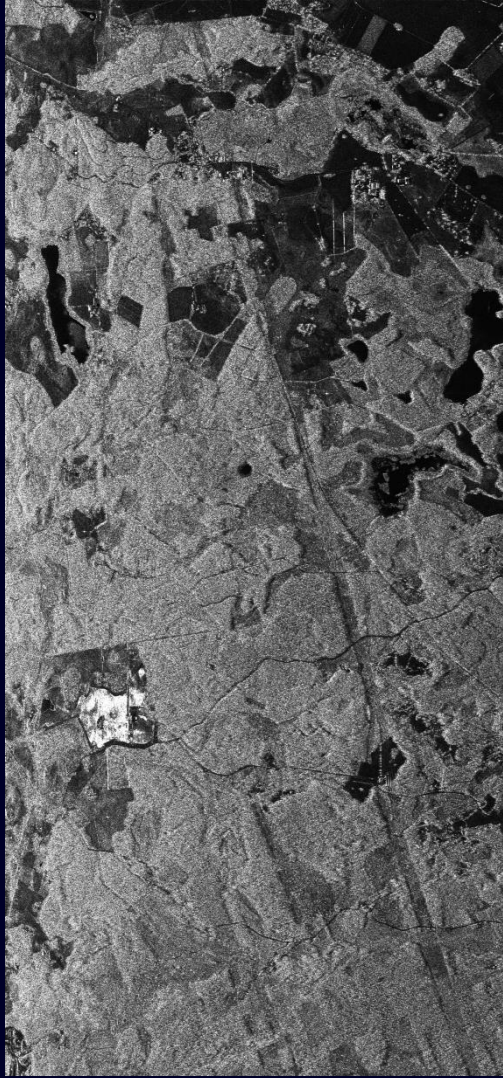
$f(z)$... vertical reflectivity function

Vertical Wavenumber: $k_z = \frac{\kappa \Delta \theta}{\sin(\theta_0)}$

Multibaseline SAR Interferometry

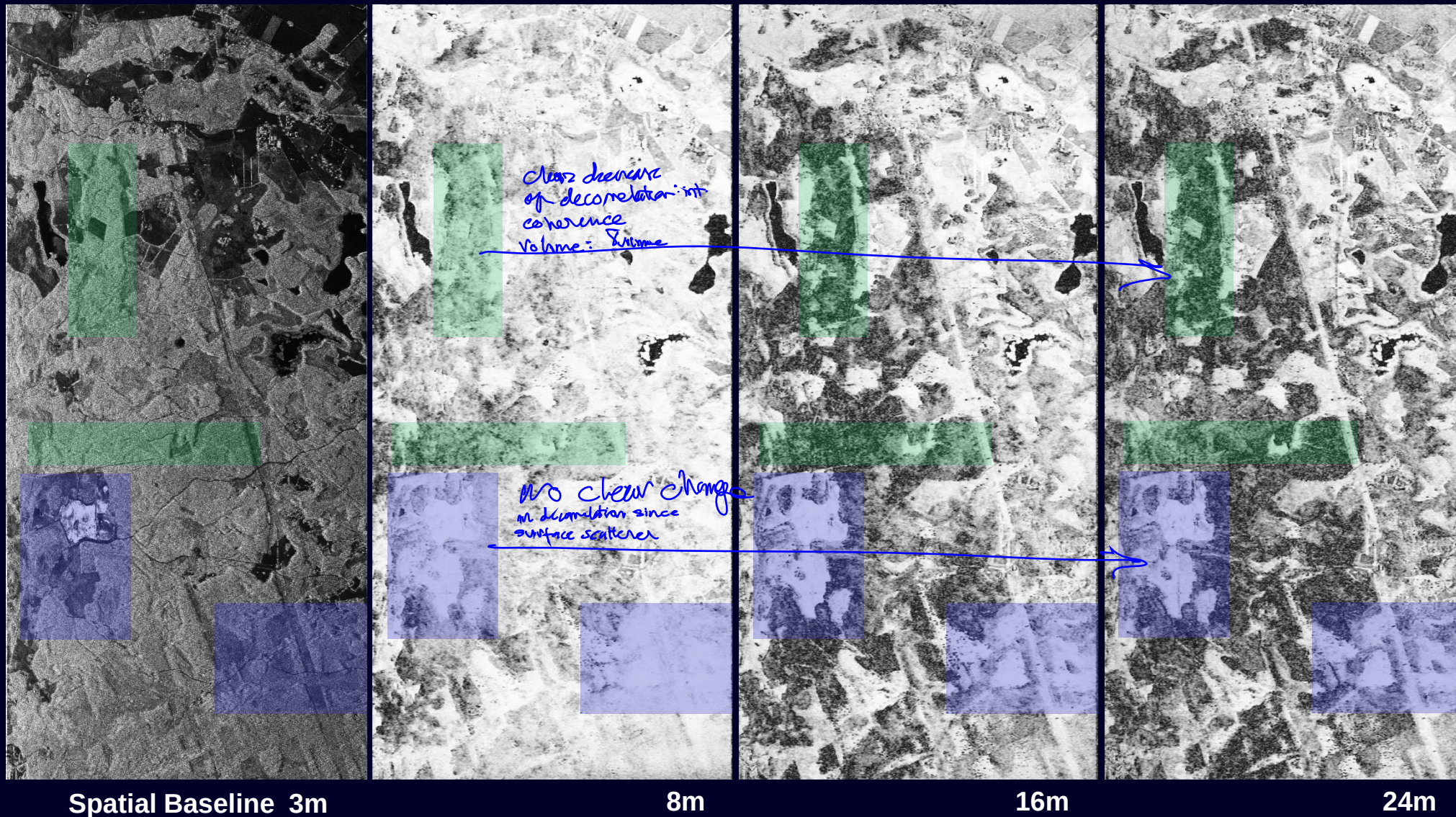


Amplitude Image



Amplitude Image HH

Interferometric Coherence: Volume Decorrelation



Polarimetric SAR Interferometry

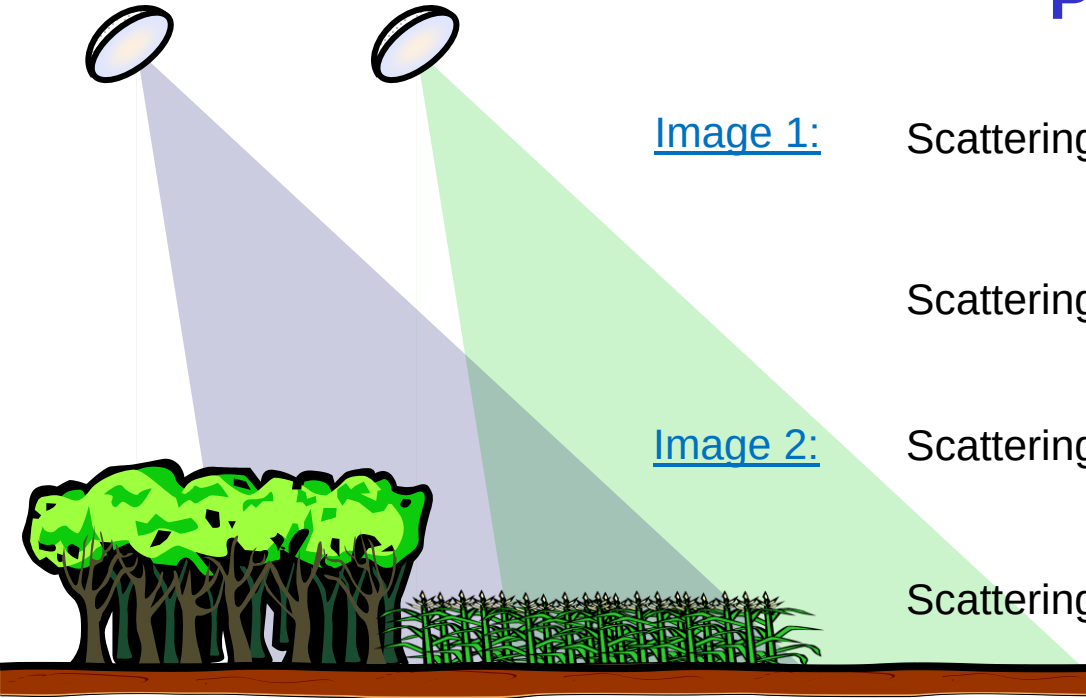


Image 1:

Scattering Matrix: $[S_1] = \begin{bmatrix} S_{HH}^1 & S_{HV}^1 \\ S_{VH}^1 & S_{VV}^1 \end{bmatrix}$

Scattering Vector: $\vec{k}_1 = \frac{1}{\sqrt{2}} [S_{HH}^1 + S_{VV}^1 \quad S_{HH}^1 - S_{VV}^1 \quad 2S_{HV}^1]^T$

Image 2:

Scattering Matrix: $[S_2] = \begin{bmatrix} S_{HH}^2 & S_{HV}^2 \\ S_{VH}^2 & S_{VV}^2 \end{bmatrix}$

Scattering Vector: $\vec{k}_2 = \frac{1}{\sqrt{2}} [S_{HH}^2 + S_{VV}^2 \quad S_{HH}^2 - S_{VV}^2 \quad 2S_{HV}^2]^T$

Image formation:

$i_1 = \vec{w}_1^+ \cdot \vec{k}_1$ and $i_2 = \vec{w}_2^+ \cdot \vec{k}_2$... projection of the scattering vector on a (complex) unitary vector $\vec{w}_i$

$\vec{w}_i$ used to select a given polarisation out of all possible polarisations provided by $[S]$

Example: $S_{HH} + S_{VV}$ image: $\vec{w} = [1 \ 0 \ 0]^T \rightarrow i = \vec{w}^+ \cdot \vec{k}_j = \frac{1}{\sqrt{2}} (S_{HH}^j + S_{VV}^j)$

S_{HH} image: $\vec{w} = [1/\sqrt{2} \ 1/\sqrt{2} \ 0]^T \rightarrow i_j = \vec{w}^+ \cdot \vec{k}_j = S_{HH}^j$

Polarimetric SAR Interferometry

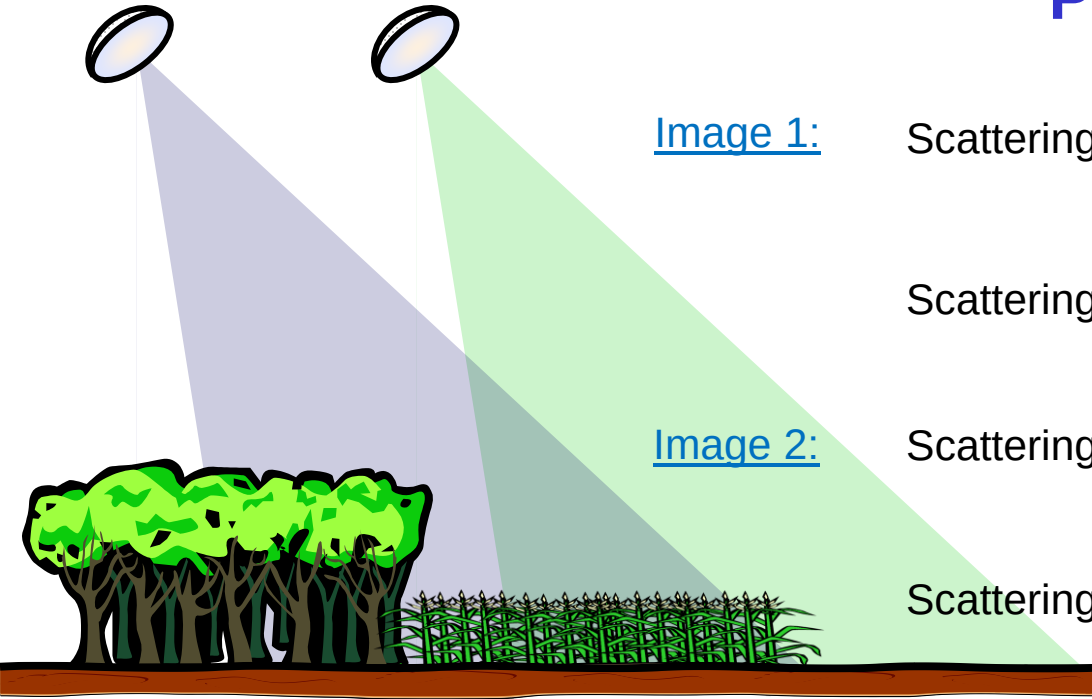


Image 1:

Scattering Matrix: $[S_1] = \begin{bmatrix} S_{HH}^1 & S_{HV}^1 \\ S_{VH}^1 & S_{VV}^1 \end{bmatrix}$

Scattering Vector: $\vec{k}_1 = \frac{1}{\sqrt{2}} [S_{HH}^1 + S_{VV}^1 \quad S_{HH}^1 - S_{VV}^1 \quad 2S_{HV}^1]^T$

Image 2:

Scattering Matrix: $[S_2] = \begin{bmatrix} S_{HH}^2 & S_{HV}^2 \\ S_{VH}^2 & S_{VV}^2 \end{bmatrix}$

Scattering Vector: $\vec{k}_2 = \frac{1}{\sqrt{2}} [S_{HH}^2 + S_{VV}^2 \quad S_{HH}^2 - S_{VV}^2 \quad 2S_{HV}^2]^T$

Image formation:

$$i_1 = \vec{w}_1^+ \cdot \vec{k}_1 \quad \text{and} \quad i_2 = \vec{w}_2^+ \cdot \vec{k}_2 \quad \text{where } \vec{w}_i \text{ are complex unitary vectors}^*$$

Interferogram formation:

$$i_1 i_2^* = (\vec{w}_1^+ \cdot \vec{k}_1)(\vec{w}_2^+ \cdot \vec{k}_2)^+ = \vec{w}_1^+ (\vec{k}_1 \cdot \vec{k}_2^+) \vec{w}_2 = \vec{w}_1^+ [\Omega] \vec{w}_2$$

Interferometric Coherence:

$$\tilde{\gamma}(\vec{w}_1, \vec{w}_2) = \frac{\langle i_1 i_2^* \rangle}{\sqrt{\langle i_1 i_1^* \rangle \langle i_2 i_2^* \rangle}} = \frac{\langle \vec{w}_1^+ [\Omega] \vec{w}_2 \rangle}{\sqrt{\langle \vec{w}_1^+ [T_{11}] \vec{w}_1 \rangle \langle \vec{w}_2^+ [T_{22}] \vec{w}_2 \rangle}}$$

where $[T_{11}] = \langle \vec{k}_1 \cdot \vec{k}_1^+ \rangle$ $[T_{22}] = \langle \vec{k}_2 \cdot \vec{k}_2^+ \rangle$ and $[\Omega] = \langle \vec{k}_1 \cdot \vec{k}_2^+ \rangle$

$\vec{w}_i$ used to select a polarisation state out of all possible polarisations provided by the scattering matrix $[S]$

Interferometric Coherence: Polarimetric Dependency



Amplitude Image HH

Sp. Baseline 16m

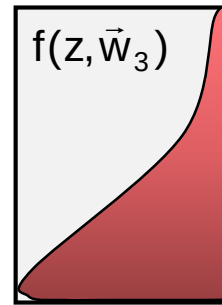
Opt 1

HH

Opt 3

Polarisation 3 ($\underline{w}_3$):

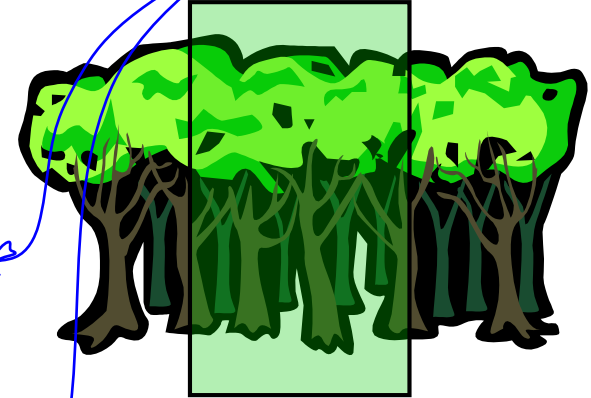
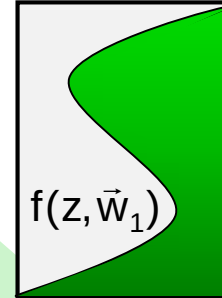
$$\tilde{\gamma}_{\text{Vol}}(f(z, \vec{w}_3)) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}_3) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}_3) dz}$$



search for the opt
in the set of $\vec{w}$'s

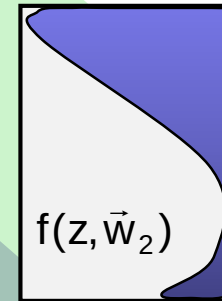
Polarisation 1 ($\underline{w}_1$):

$$\tilde{\gamma}_{\text{Vol}}(f(z, \vec{w}_1)) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}_1) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}_1) dz}$$



Polarisation 2 ($\underline{w}_2$):

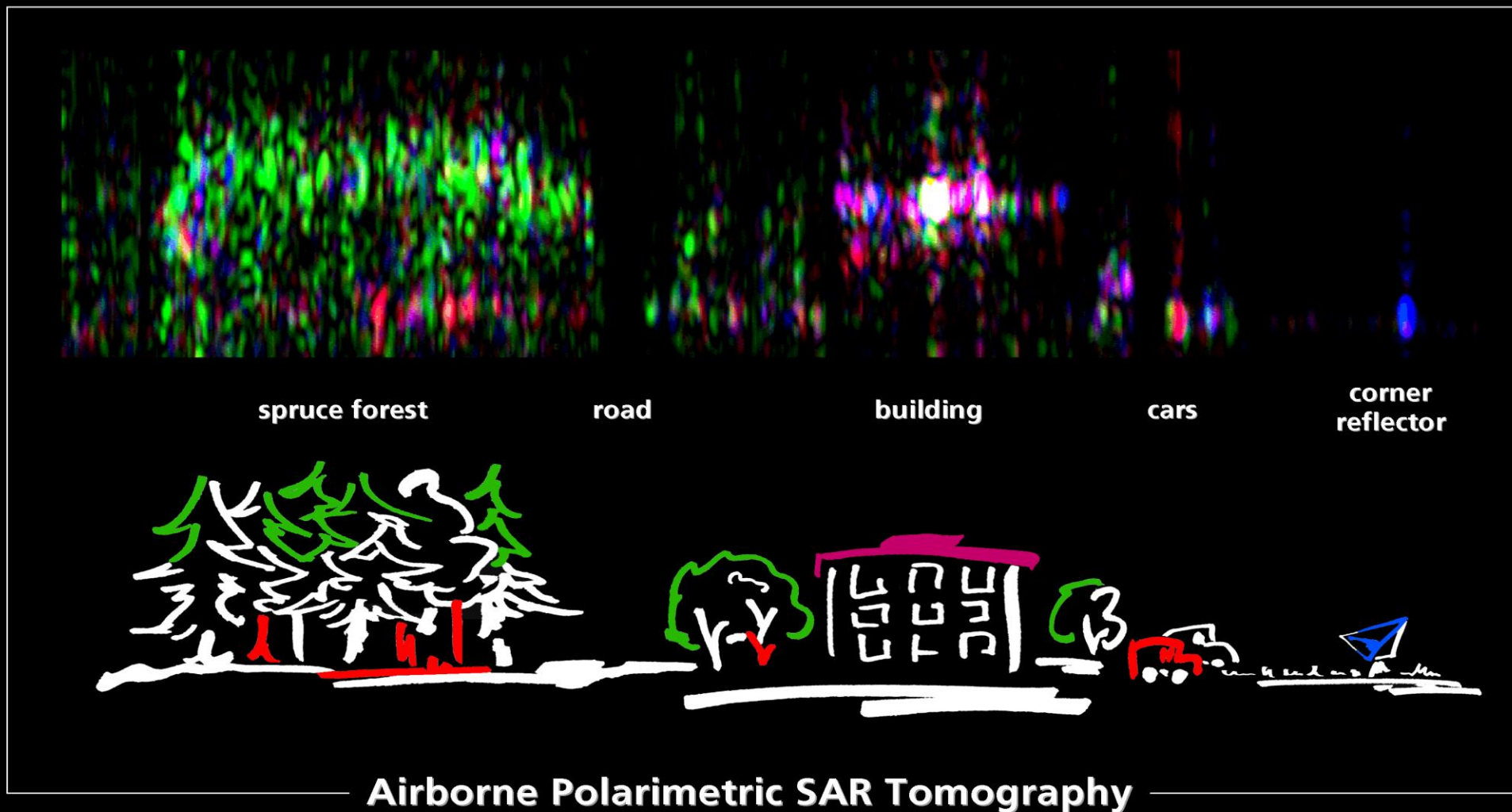
$$\tilde{\gamma}_{\text{Vol}}(f(z, \vec{w}_2)) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}_2) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}_2) dz}$$



$f(z, \vec{w})$...vertical reflectivity function

Polarimetric SAR Interferometry

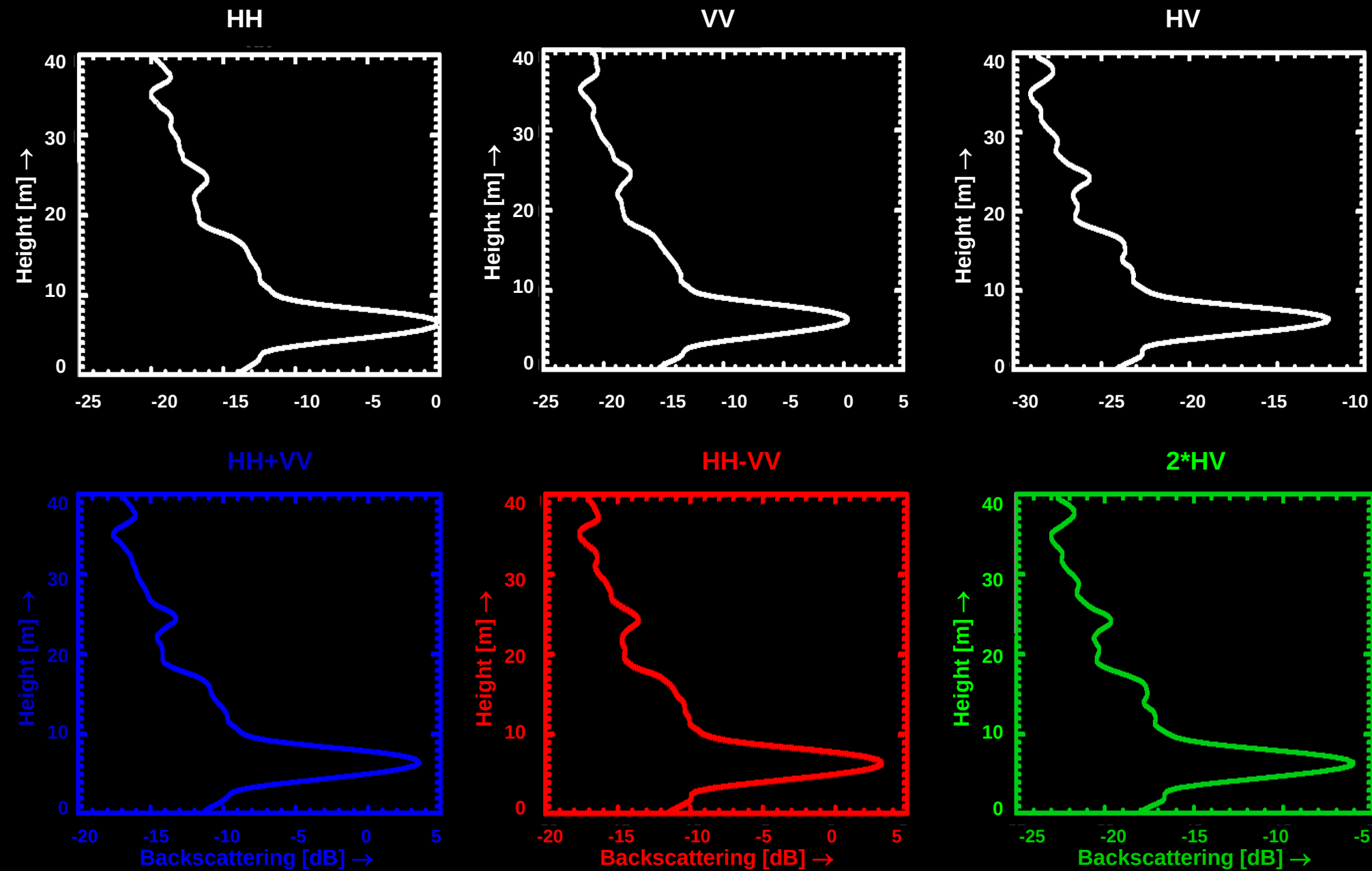




Upper image: Polarimetric color composite (L-band) of a tomographic slice in the height/azimuth-direction

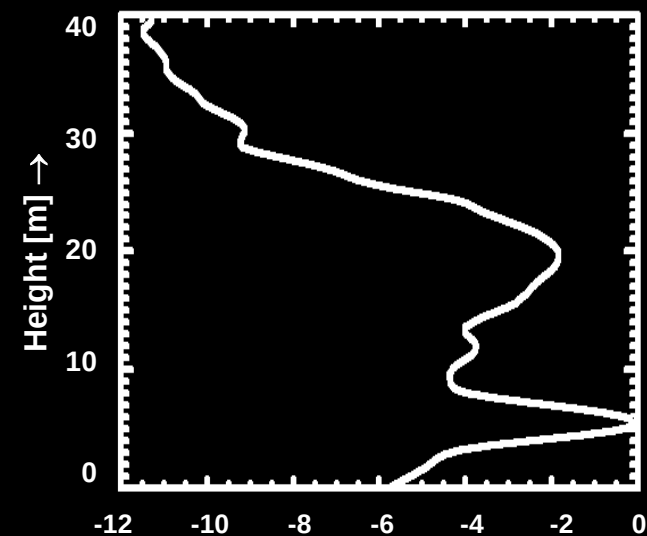
■ HH+VV, ■ HH-VV, ■ 2*HV

Lower image: Schematic view of the imaged area

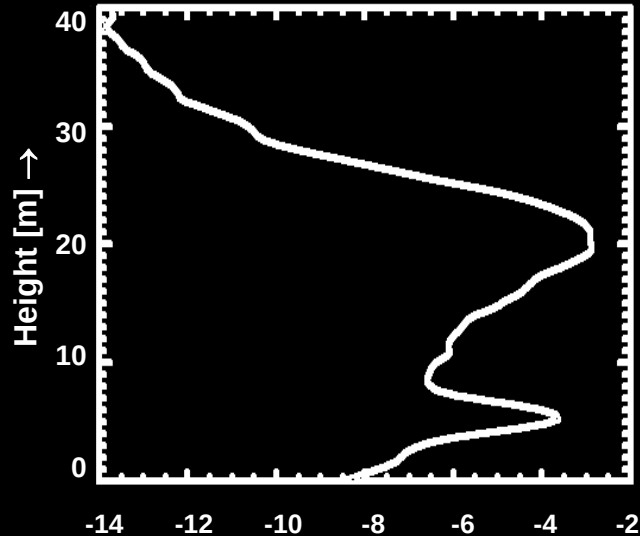


Bare Surface Backscattering Profiles (12-20 m height)

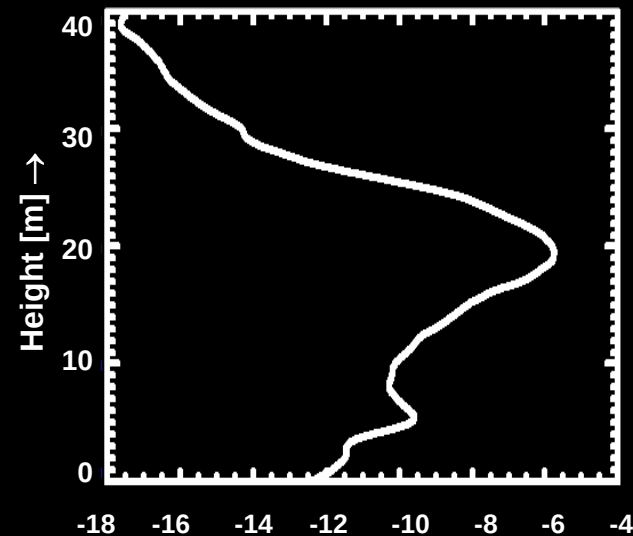
HH



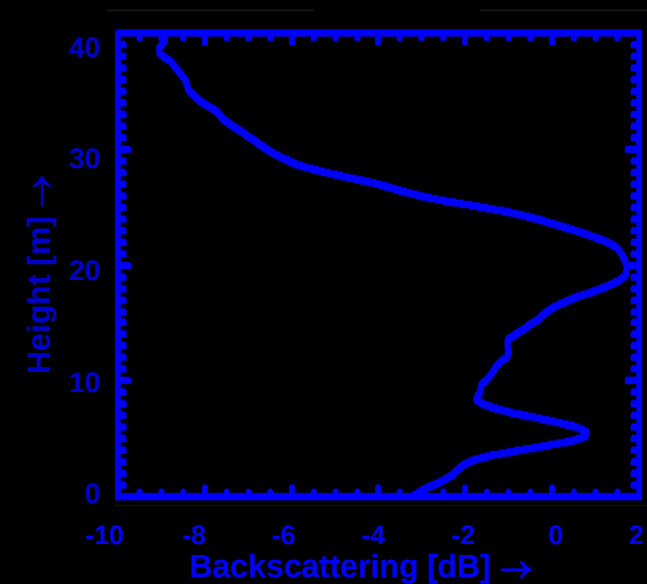
VV



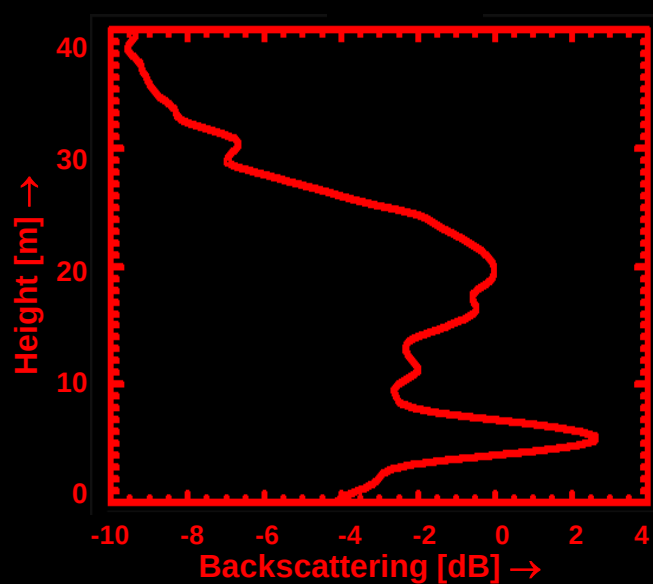
HV



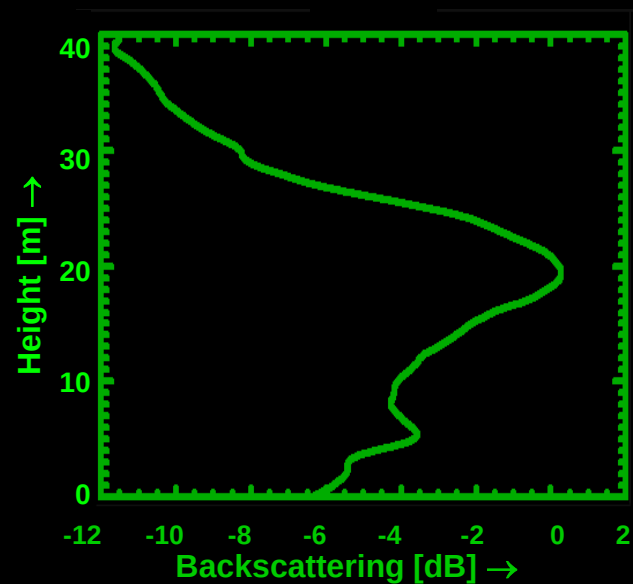
HH+VV



HH-VV

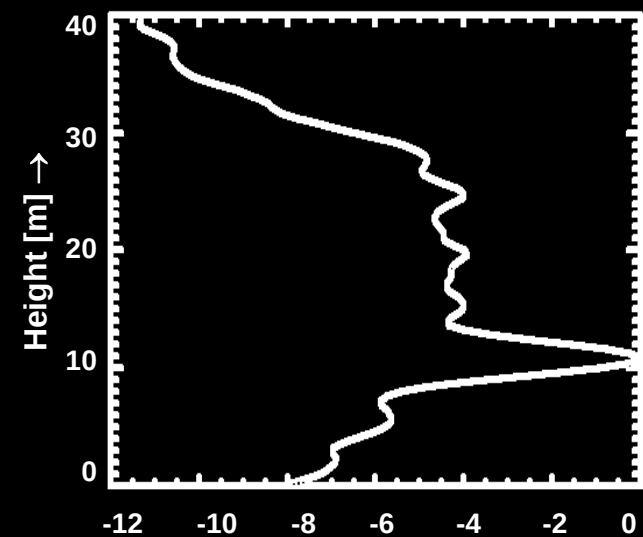


2*HV

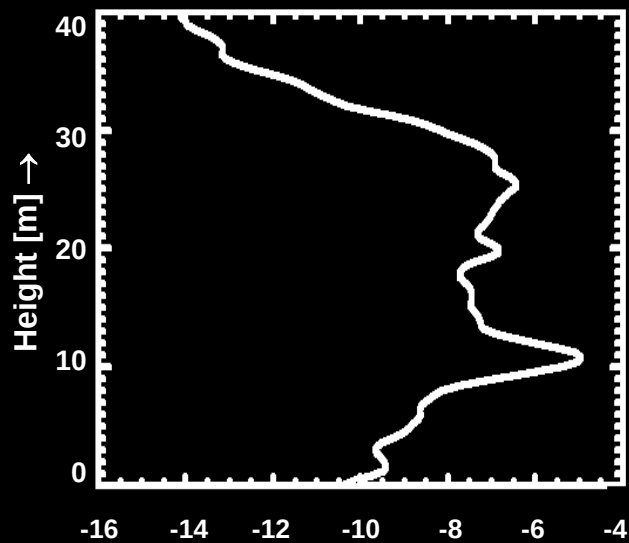


Spruce Forest Backscattering Profiles (15-20 m height)

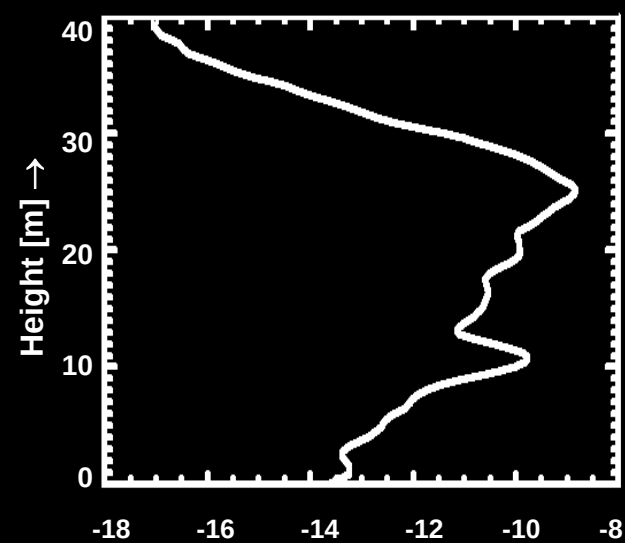
HH



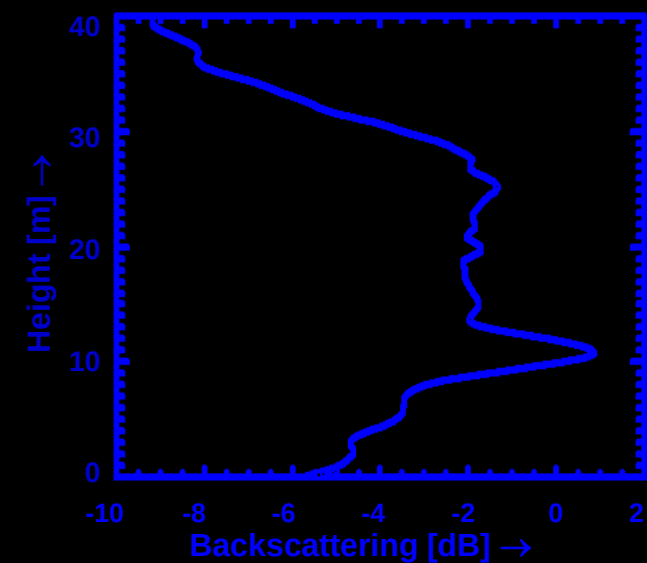
VV



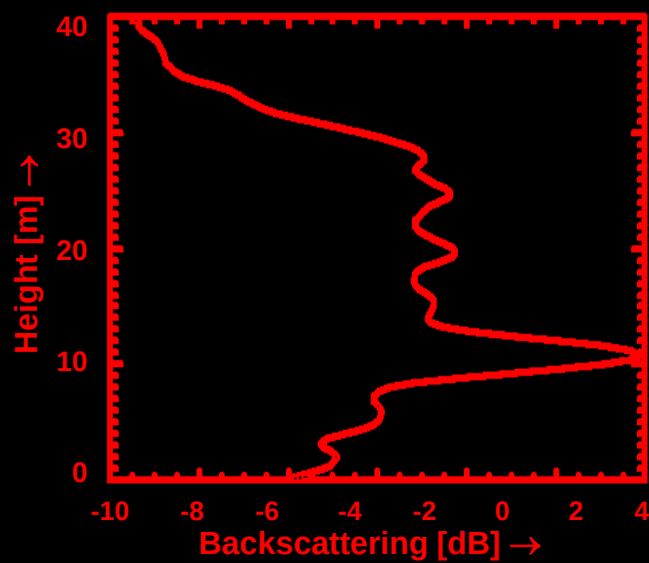
HV



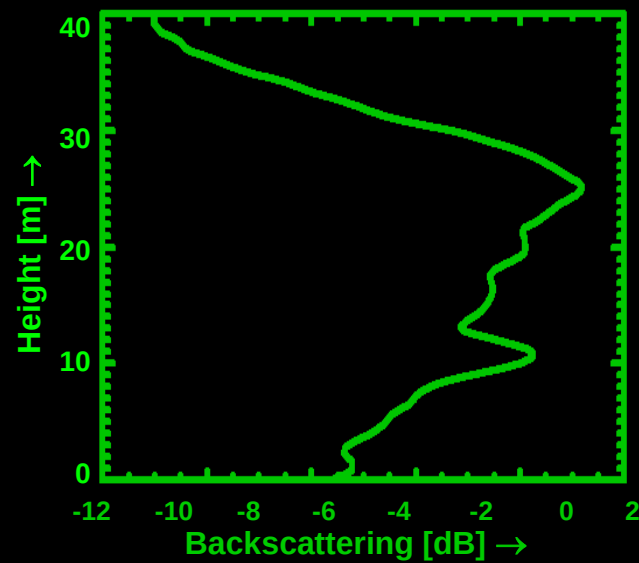
HH+VV



HH-VV



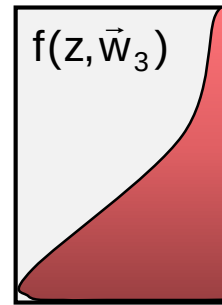
2*HV



Mixed Forest Backscattering Profiles (12-20 m height)

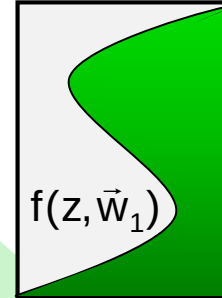
Polarisation 3 ($\vec{w}_3$):

$$\tilde{\gamma}_{\text{Vol}}(f(z, \vec{w}_3)) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}_3) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}_3) dz}$$



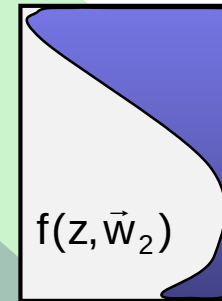
Polarisation 1 ($\vec{w}_1$):

$$\tilde{\gamma}_{\text{Vol}}(f(z, \vec{w}_1)) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}_1) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}_1) dz}$$



Polarisation 2 ($\vec{w}_2$):

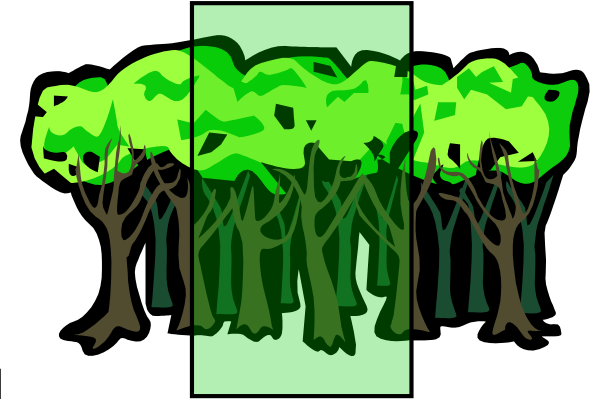
$$\tilde{\gamma}_{\text{Vol}}(f(z, \vec{w}_2)) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z, \vec{w}_2) e^{ik_z z} dz}{\int_0^{h_v} f(z, \vec{w}_2) dz}$$



$f(z, \vec{w})$...vertical reflectivity function

Polarimetric SAR Interferometry

By changing the polarization, the contrast between the individual components which form the vertical reflectivity $f(z)$ of a (volume) scatterer changes.

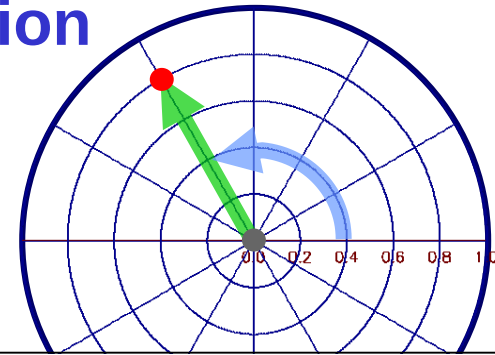


Geometrical Representation: The Coherence Region

Interferometric Coherence: $\tilde{\gamma}(\vec{w}_i, \vec{w}_i) = |\tilde{\gamma}(\vec{w}_i, \vec{w}_i)| \cdot \exp(i \text{Arg}\{\tilde{\gamma}(\vec{w}_i, \vec{w}_i)\})$

Radius

Angle



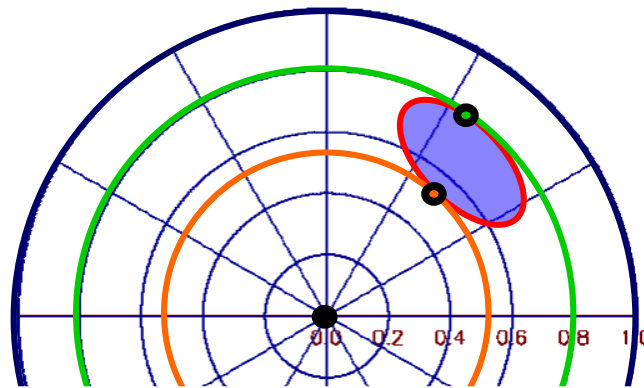
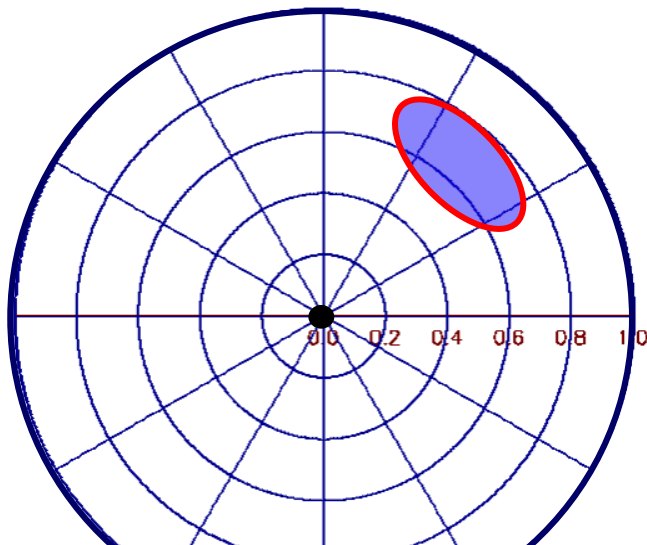
► can be represented by a single point on the unit circle (UC)

Coherence Region:

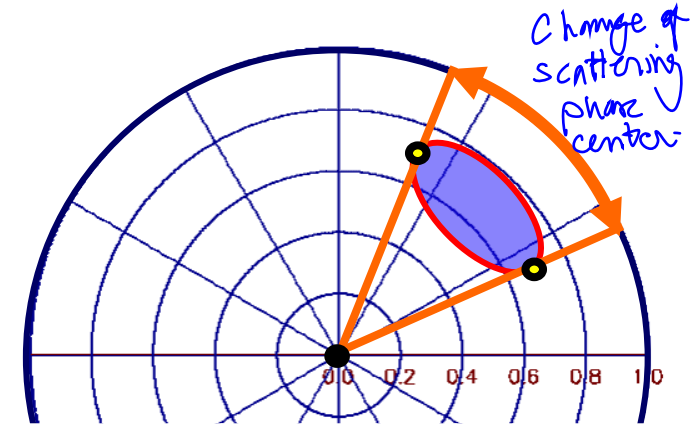
$$\tilde{\gamma}(\vec{w}_i, \vec{w}_i) \quad \forall \quad \vec{w}_i = \begin{bmatrix} \cos \alpha \exp(i\phi_1) \\ \sin \alpha \cos \beta \exp(i\phi_2) \\ \sin \alpha \sin \beta \exp(i\phi_3) \end{bmatrix} \quad \text{with} \quad \underbrace{0 \leq \alpha \leq \frac{\pi}{2}}_{\text{type of scatterer}} \quad \text{and} \quad \underbrace{-\pi \leq \beta \leq \pi}_{\text{LOS rotation or scatterer w.r. radar}}$$

The set of interferometric coherences obtained for all the possible polarizations $\vec{w}_i$ plotted on the unit circle (UC) defines the so-called **coherence region**.

Its shape & size depend on the structure of the underlying scatterer and on acquisition parameters.

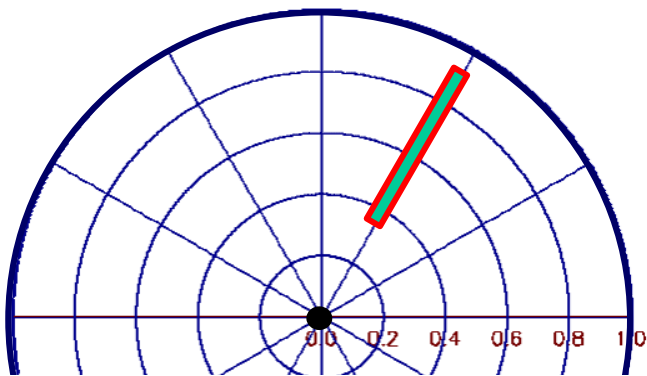


Max./ Min. Interferometric Coherence
as function of $\vec{w}_i$



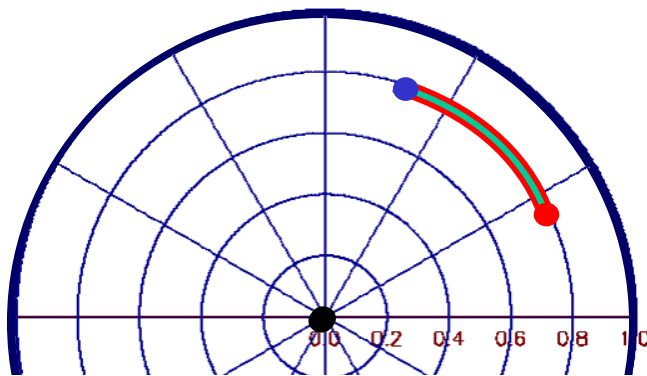
Max. Phase Difference between
interferograms formed with $\vec{w}_i$ and $\vec{w}_j$

Coherence Region (CR) Interpretation



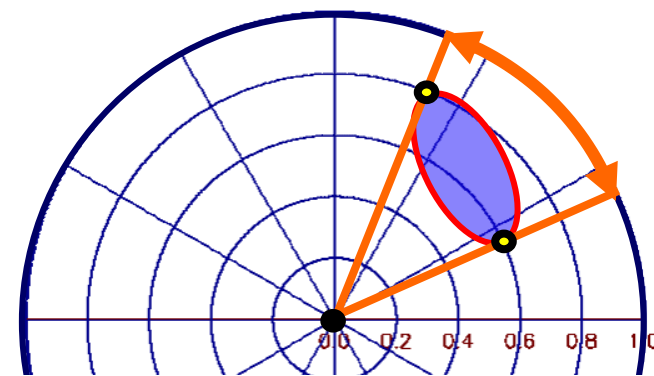
Radial Shaped CR

i.e. InSAR coherence amplitude changes with polarisation but not the location of the phase center.



Arc Shaped CR

i.e. InSAR phase center location changes with polarisation but not the absolute value of the coherence amplitude.

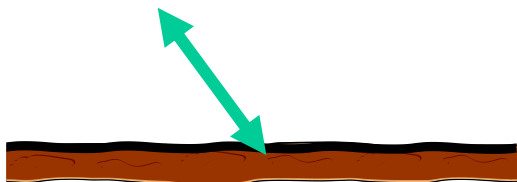


Elliptical Shaped CR

i.e. InSAR coherence magnitude and phase center location changes with polarisation.

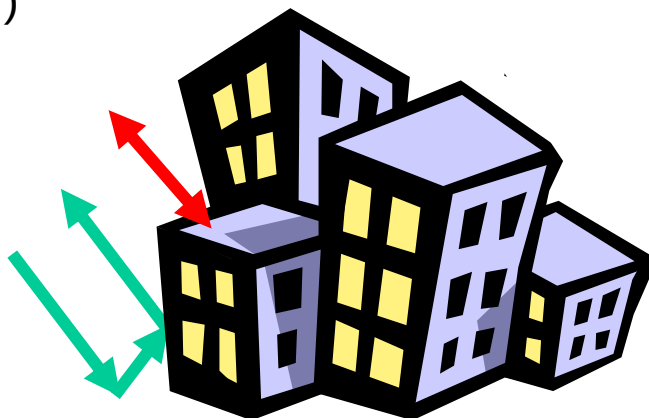
Surface Scattering

$$\tilde{Y}(\vec{W}) = Y_{\text{SNR}}(\vec{W}) \quad \tilde{Y}_{\text{Vol}}^{\tilde{Y}_{\text{Vol}}:=1} = Y_{\text{SNR}}(\vec{W})$$



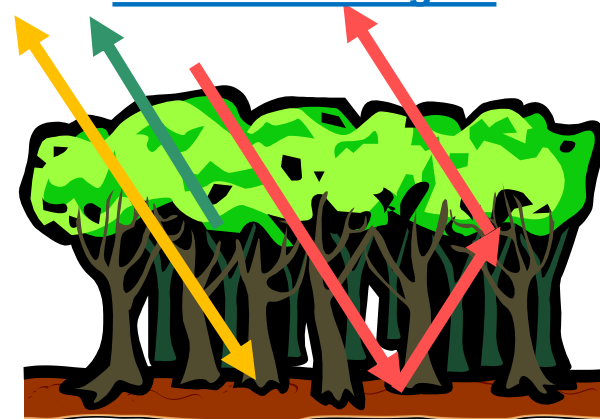
(Polarised) Coherent scatterers

at different heights

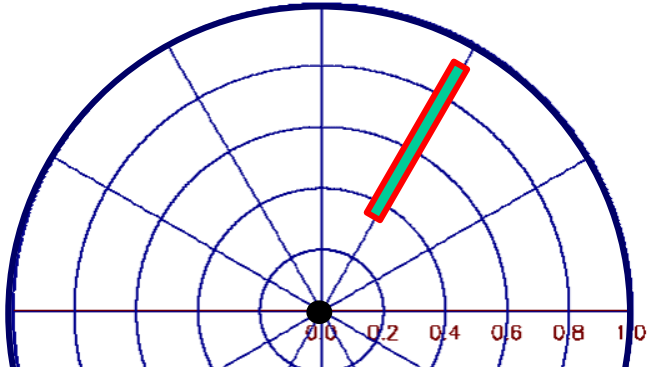


(Depolarising) Scatterers

at different heights

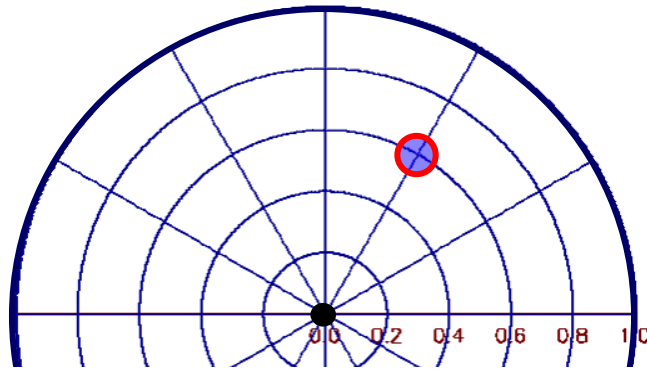


Coherence Region Interpretation



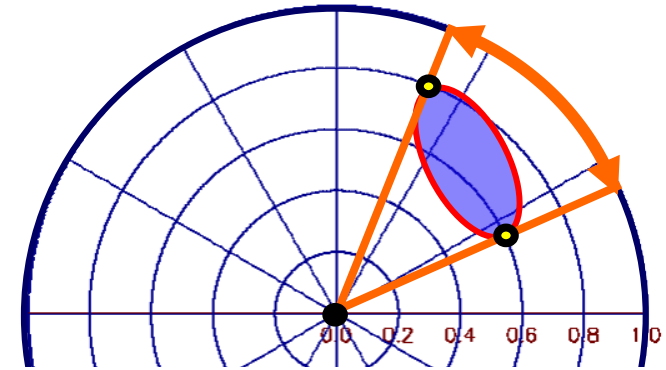
Radial Shaped CR

i.e. InSAR coherence amplitude changes with polarisation but not the location of the phase center.



Point Like Coherence Region

i.e. InSAR Coherence and Phase are independent of polarisation.

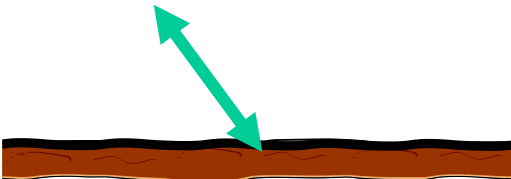


Elliptical Shaped CR

i.e. InSAR coherence magnitude and phase center location changes with polarisation.

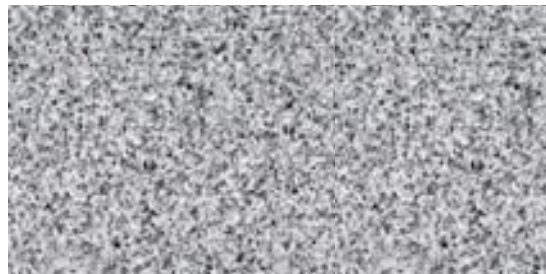
Surface Scattering

$$\tilde{Y}(\vec{W}) = Y_{\text{SNR}}(\vec{W}) \quad \tilde{Y}_{\text{Vol}}^{\tilde{Y}_{\text{Vol}}:=1} = Y_{\text{SNR}}(\vec{W})$$

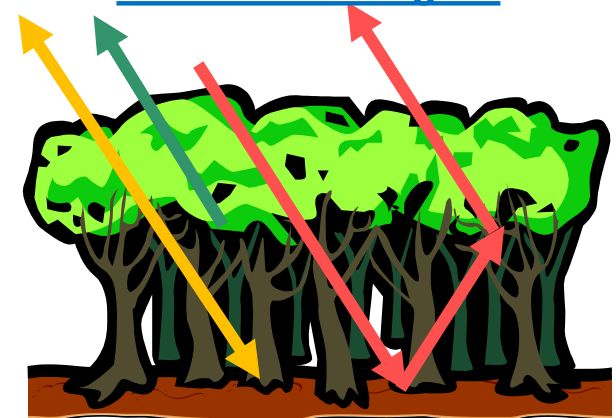


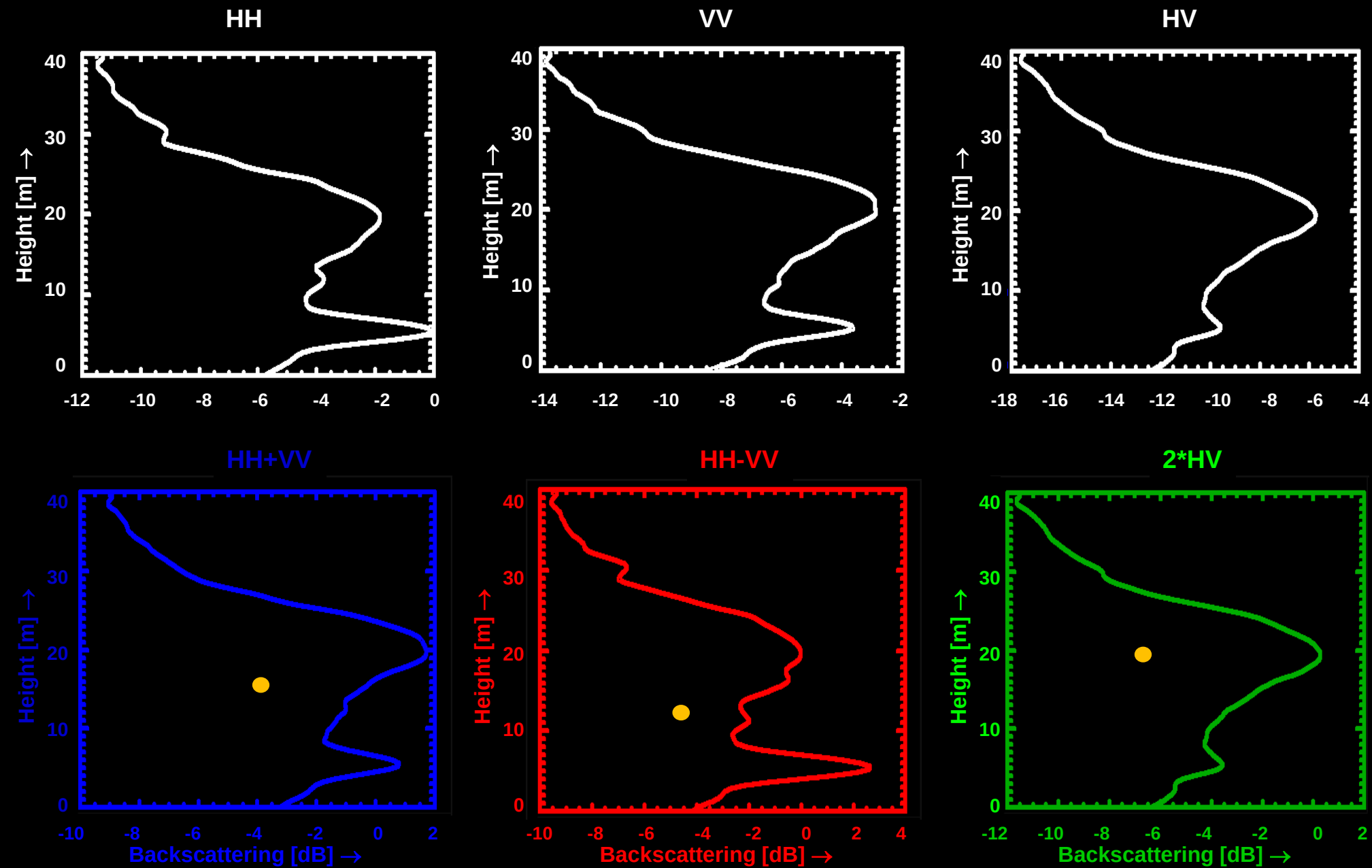
Pol-InSAR does not provide any additional information compared to InSAR !!!

(Random) Volume scattering



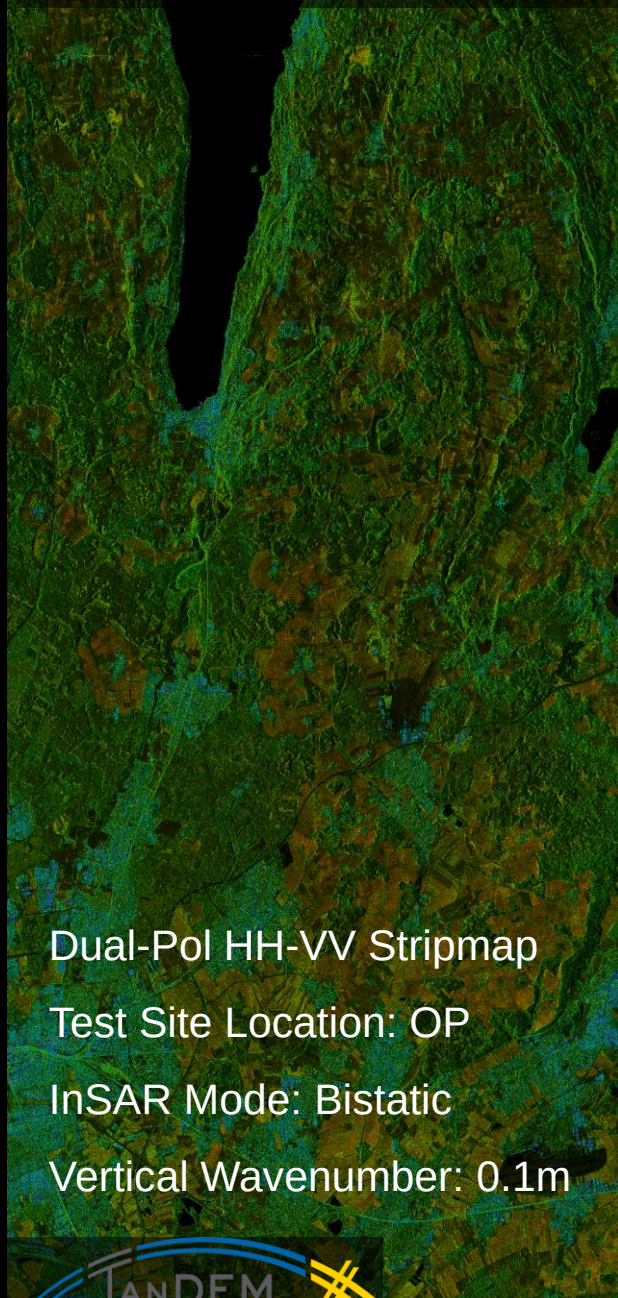
(Depolarising) Scaterrers at different heights





Spruce Forest Backscattering Profiles (15-20 m height)

First Bistatic Dual Pol-InSAR Data Takes



Dual-Pol HH-VV Stripmap

Test Site Location: OP

InSAR Mode: Bistatic

Vertical Wavenumber: 0.1m

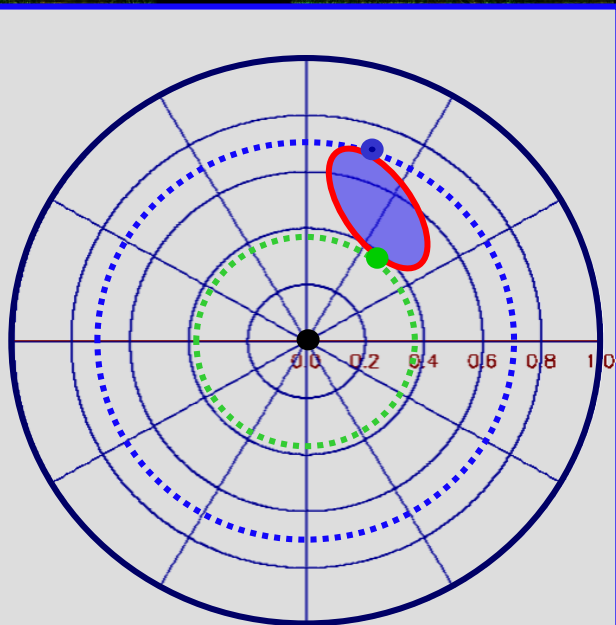


HH



VV

First Bistatic Dual Pol-InSAR Data Takes



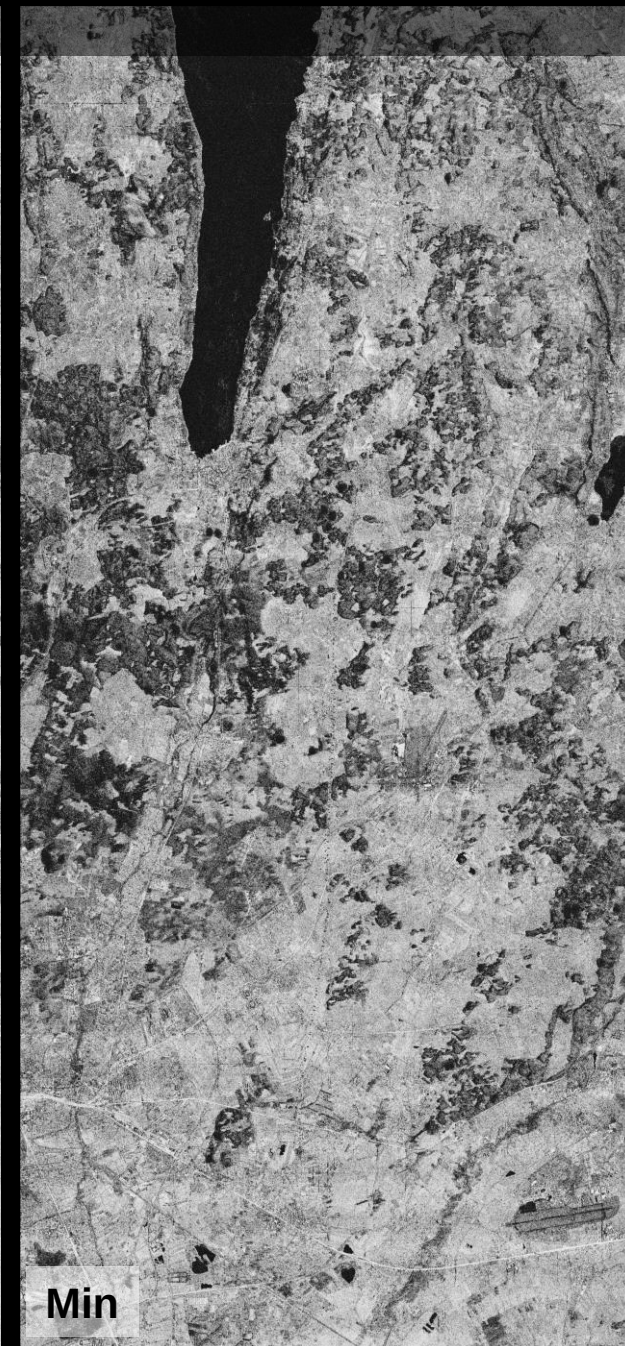
Max. Coherence Difference
between Polarizations

Dual-Pol HH-VV Stripmap

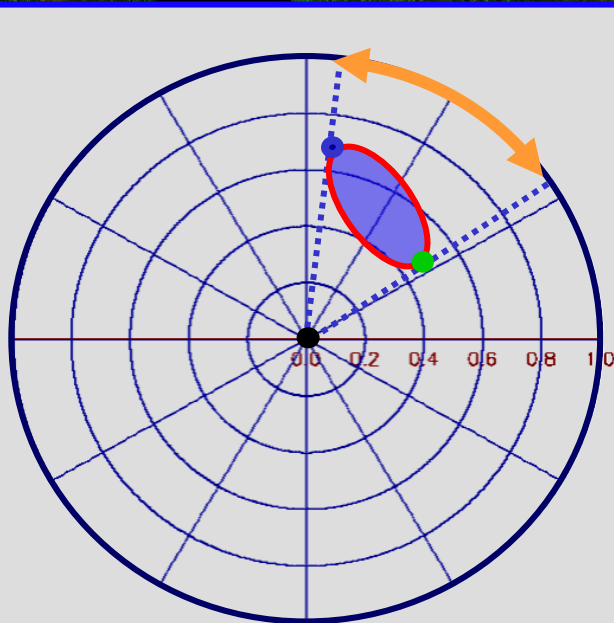
Test Site Location: OP

InSAR Mode: Bistatic

Vertical Wavenumber: 0.1m



First Bistatic Dual Pol-InSAR Data Takes



Max. Phase Difference
between Polarizations

Dual-Pol HH-VV Stripmap

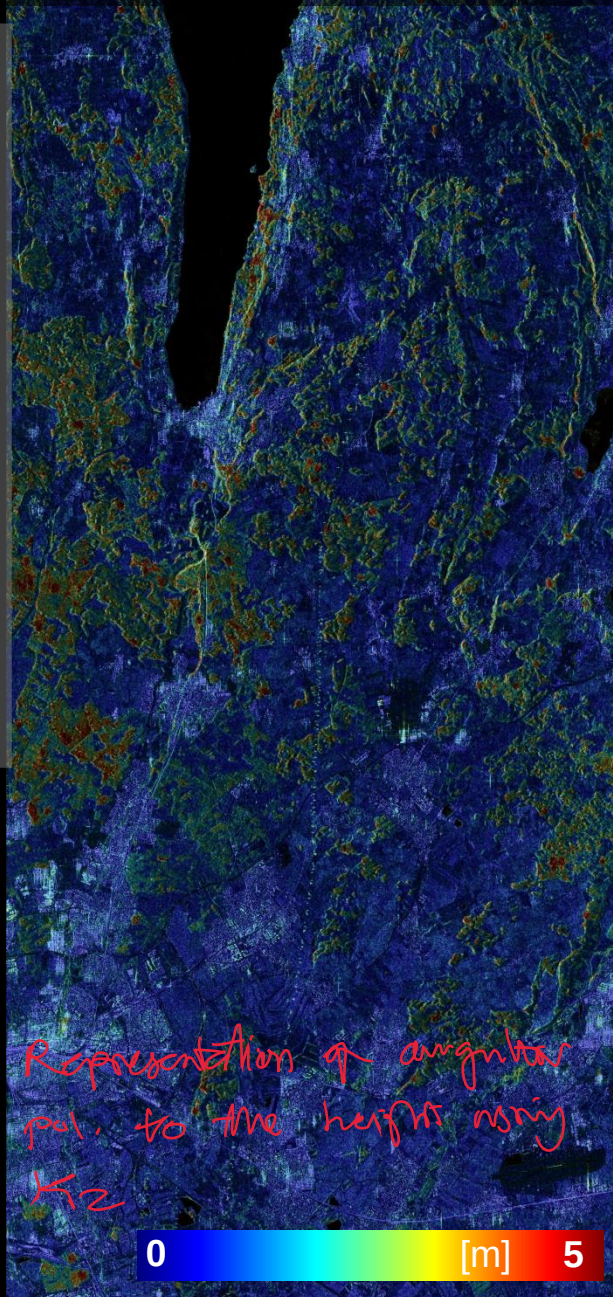
Test Site Location: OP

InSAR Mode: Bistatic

Vertical Wavenumber: 0.1m

*Representation of angular
rot. to the height using
 K_2*

0 [m] 5



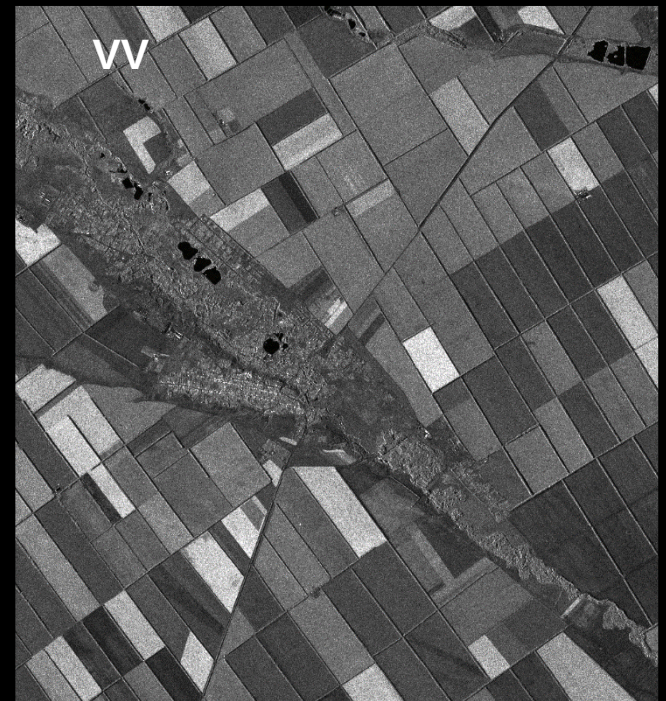
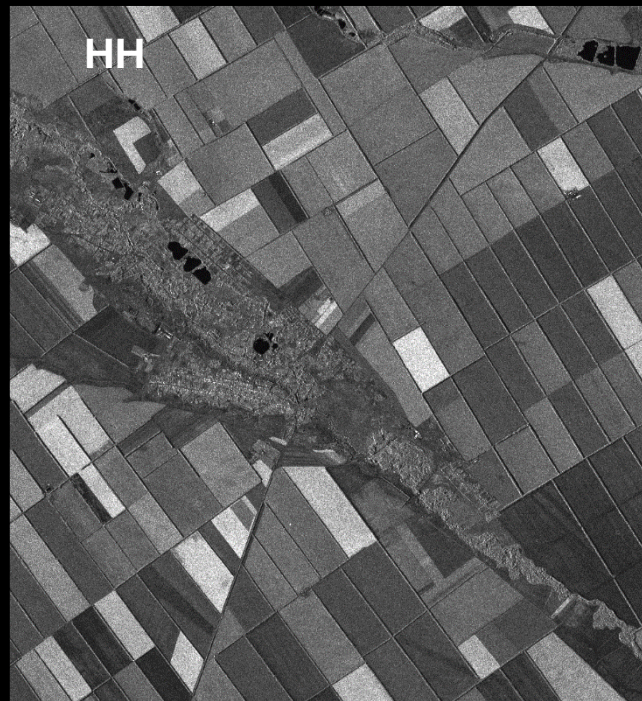
TanDEM-X Pol-InSAR Data Takes

Dual-Pol HH-VV Spotlight
Test Site Location: Russia
InSAR Mode: Monostatic
Temporal Baseline: 3sec
Spatial Baseline ($\perp$): 275m



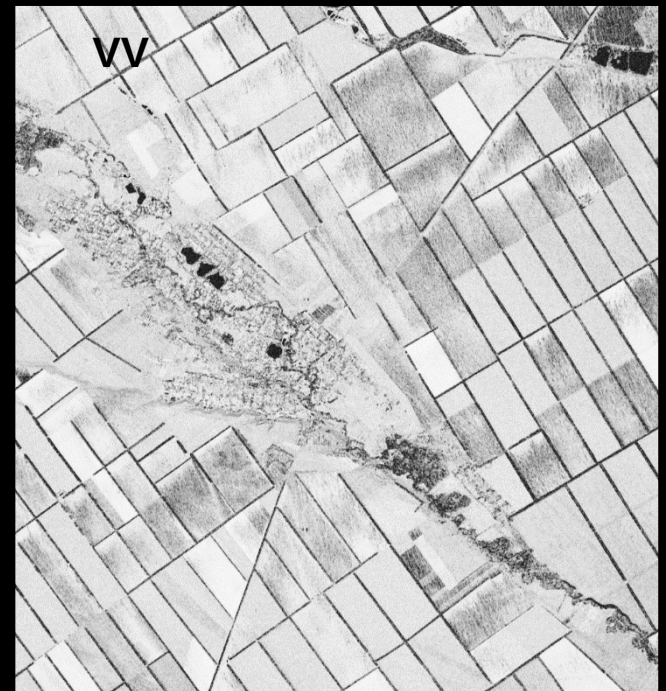
polarizations

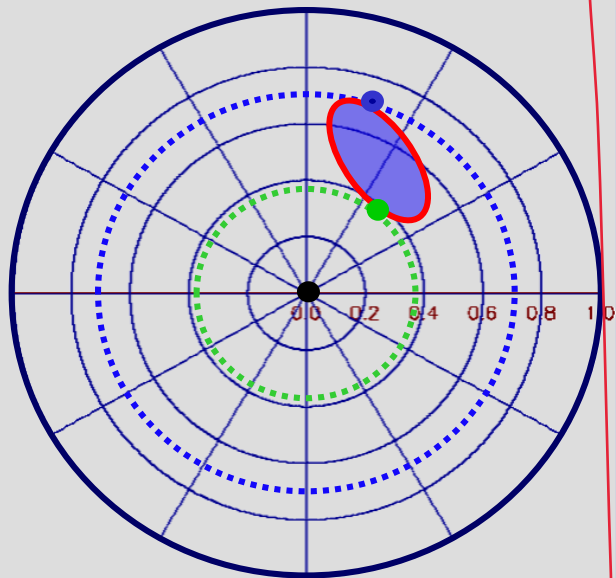
Amplitude Images



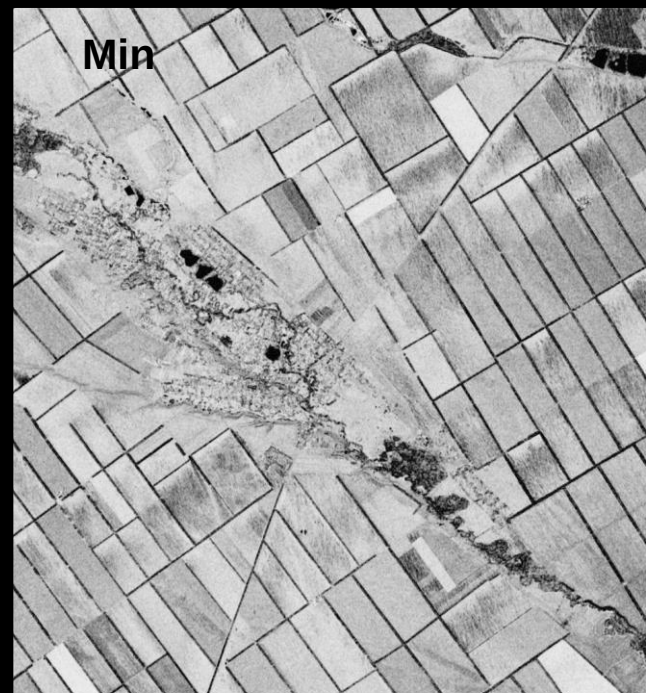
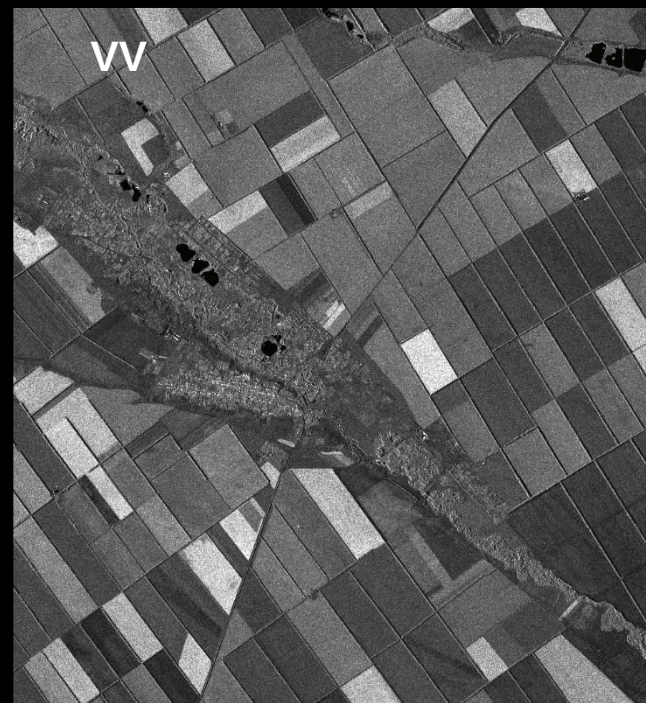
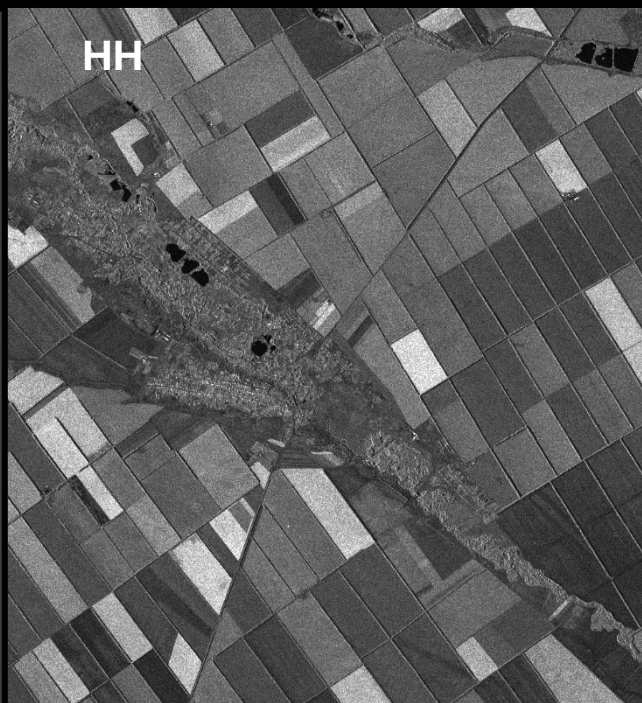
Interferometric Coherence

coherency

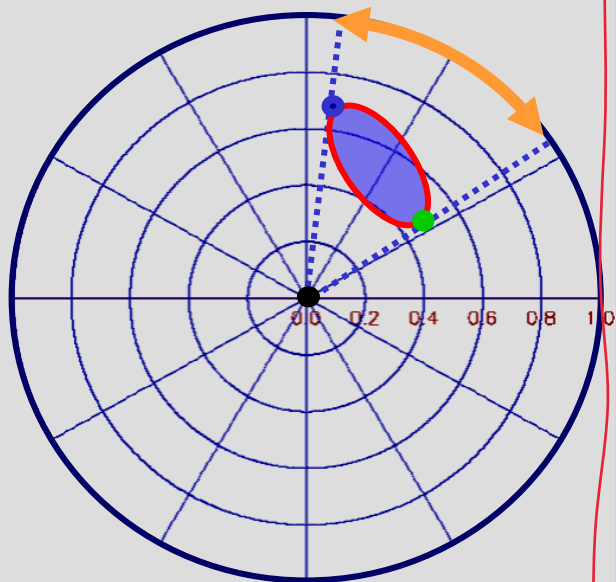




Max. Coherence Difference
between Polarisations



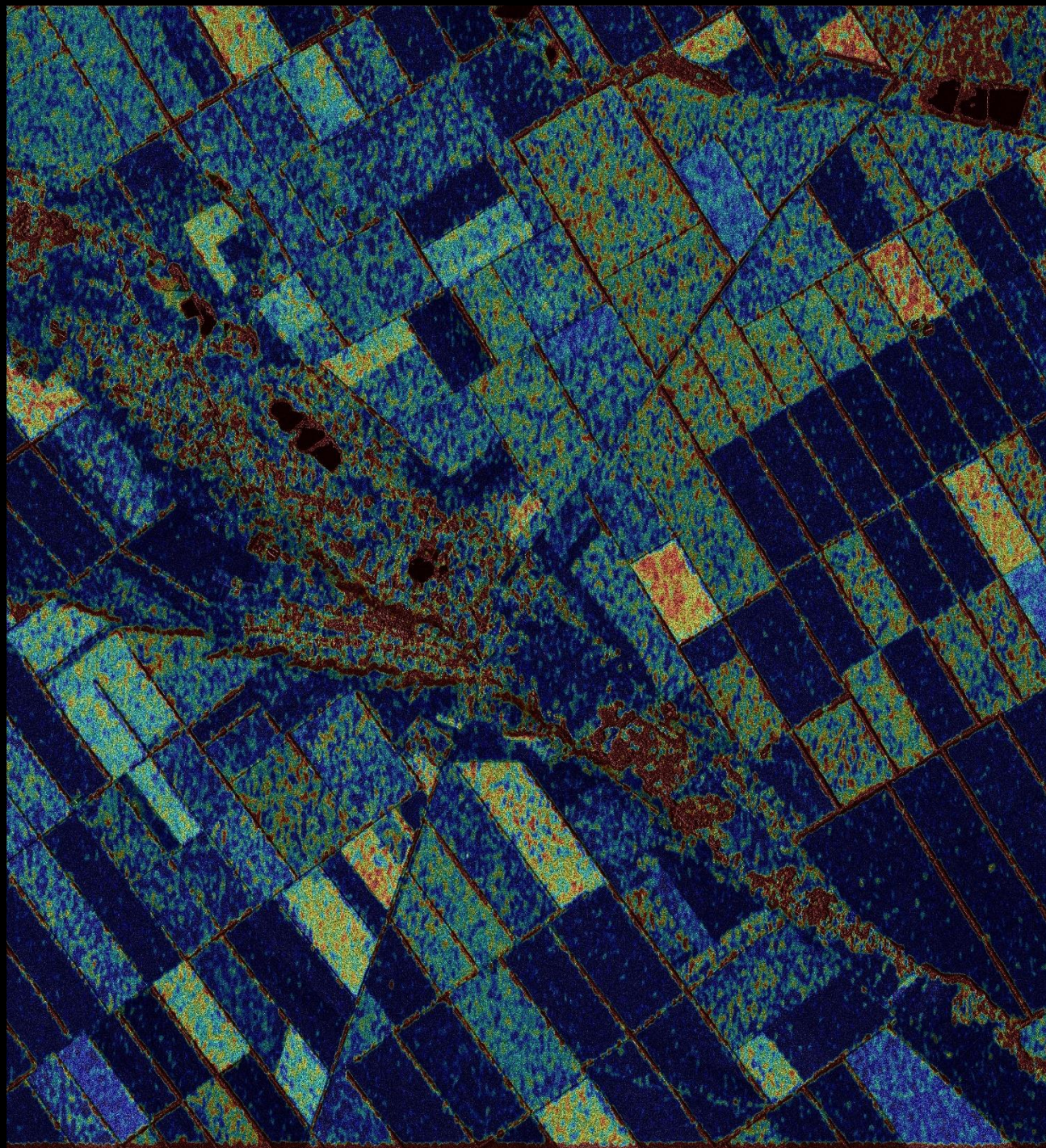
Use
pol InSAR

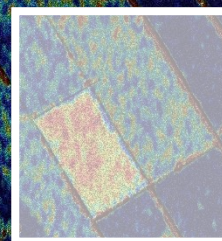
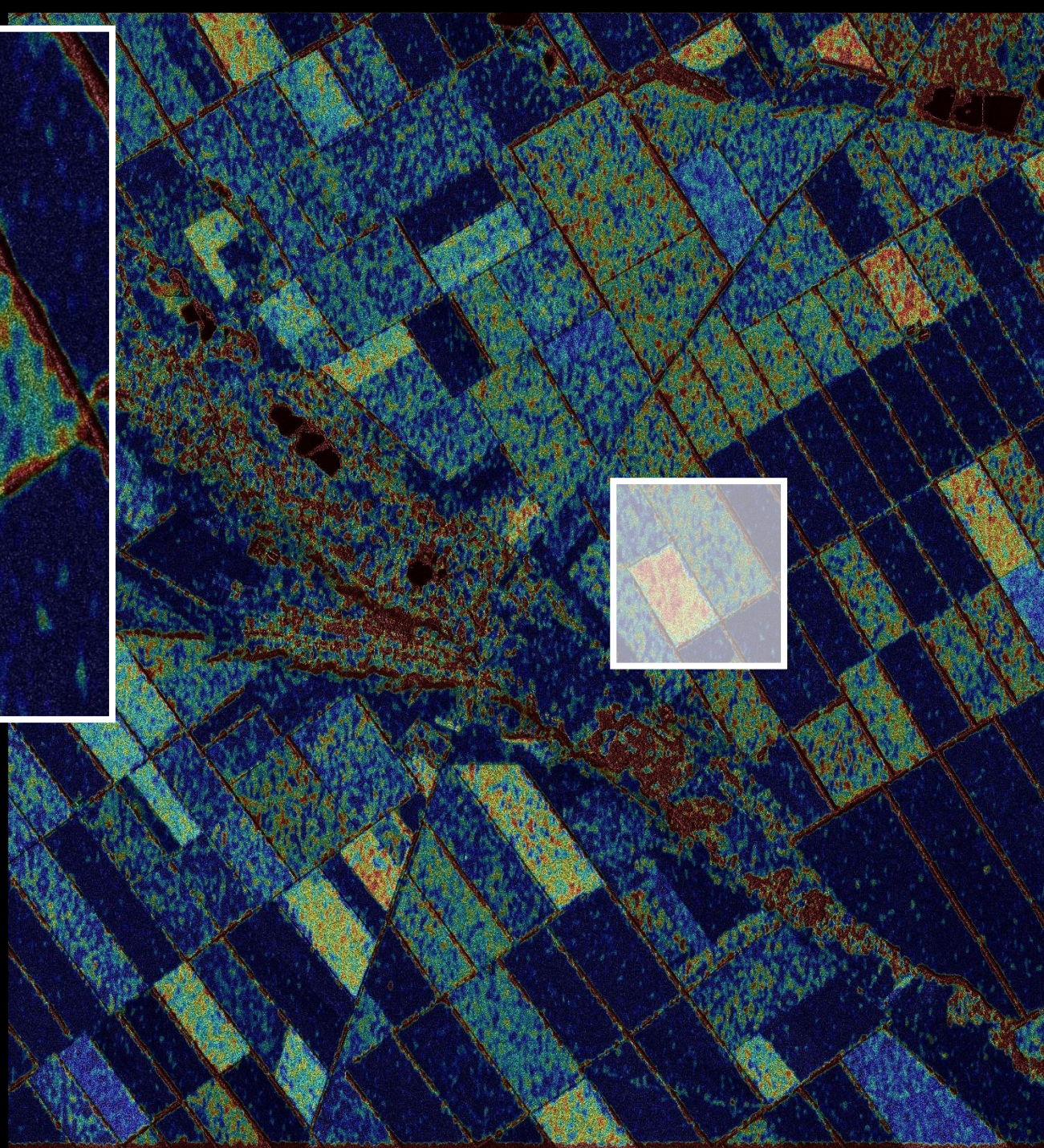
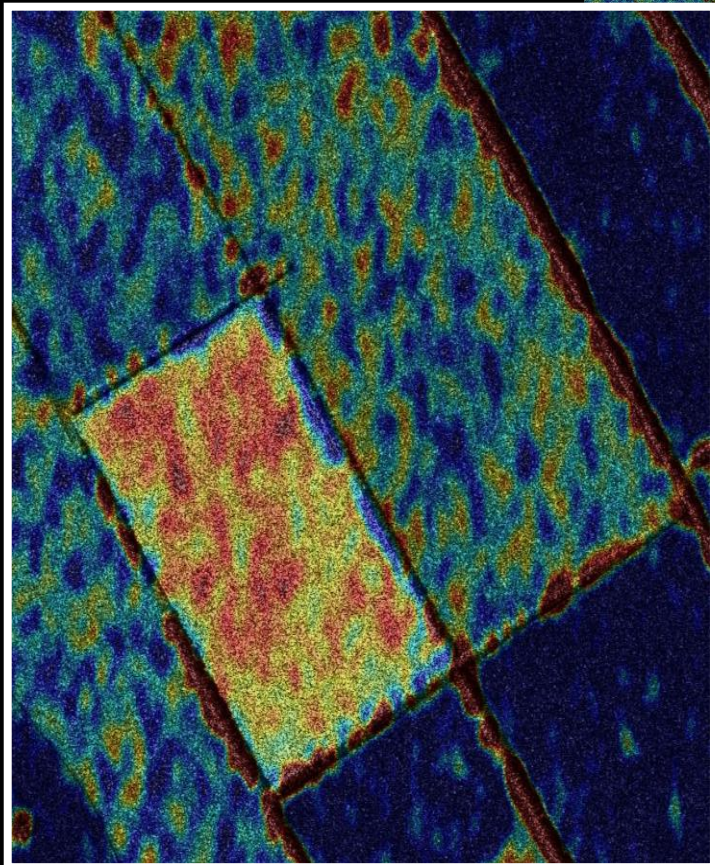


Max. Phase Difference
between Polarisations

use pol.
check

0 1 2 3 [m]





0 1 2 [m] 3

The Pol-InSAR Course

Part 8: Pol-InSAR 1

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

Email: noelia.romeropuig@dlr.de, matteo.pardini@dlr.de, kostas.papathanassiou@dlr.de



ESAMAAP

EEBIOMASS

Jupyter Notebook: Pol-InSAR Part 1 - Workflow

- **Input parameters**
 - **Step 1:** Open images, and visualize a Pauli
 - **Step 2:** Compensate flat-earth
 - **Step 3:** Calculate the elements of T1, T2, and Omega matrices
 - **Step 4:** Pre-whitening (normalization)
 - **Step 5:** Visualize the Coherence Region
-
- **Step 6:** Calculate the geometry of the Coherence Region
 - **Step 7:** Calculate the extreme coherences
 - **Step 8:** Visualize the Coherence Region + computed coherences
 - **Step 9:** Plots!

Polarimetric SAR Interferometry

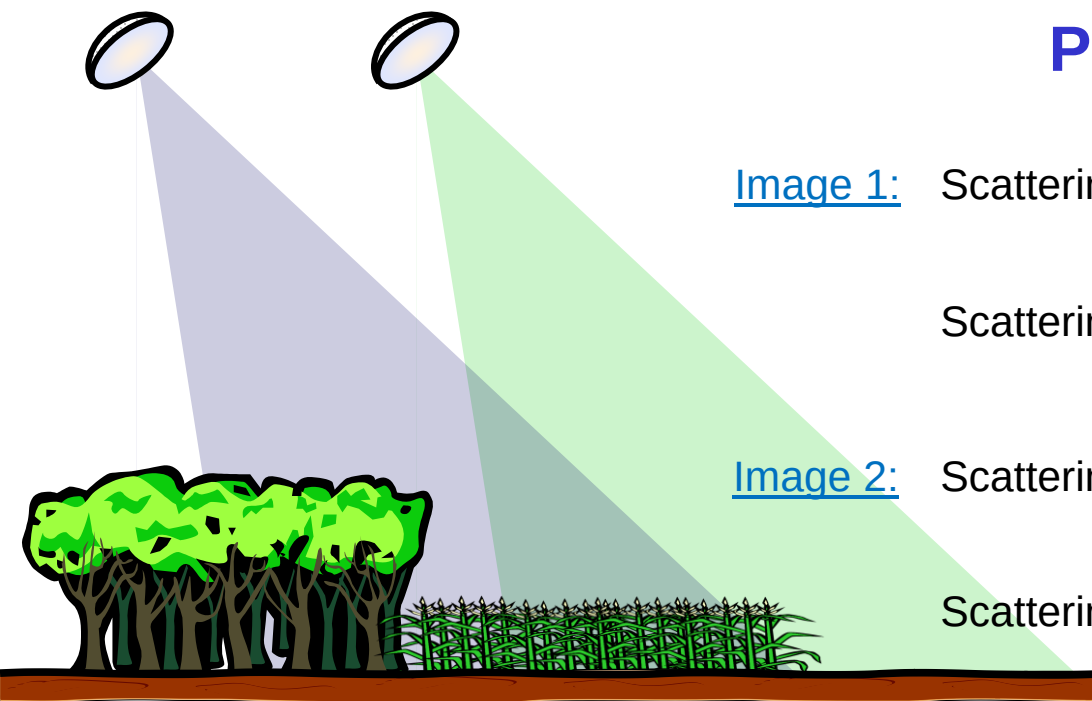


Image 1: Scattering Matrix: $[S_1] = \begin{bmatrix} S_{HH}^1 & S_{HV}^1 \\ S_{VH}^1 & S_{VV}^1 \end{bmatrix}$

Scattering Vector: $\vec{k}_1 = \frac{1}{\sqrt{2}} [S_{HH}^1 + S_{VV}^1 \quad S_{HH}^1 - S_{VV}^1 \quad 2S_{HV}^1]^T$

Image 2: Scattering Matrix: $[S_2] = \begin{bmatrix} S_{HH}^2 & S_{HV}^2 \\ S_{VH}^2 & S_{VV}^2 \end{bmatrix}$

Scattering Vector: $\vec{k}_2 = \frac{1}{\sqrt{2}} [S_{HH}^2 + S_{VV}^2 \quad S_{HH}^2 - S_{VV}^2 \quad 2S_{HV}^2]^T$

1. Pol-InSAR Matrices: The Dual-Pol Case

A: Polarisation 1, B: Polarisation 2

- Two scattering vectors $\vec{k}_1$ and $\vec{k}_2$ from $[S_1]$ and $[S_2]$: $\vec{k}_1 = \frac{1}{\sqrt{2}} [S_A^1 \quad S_B^1]^T$ and $\vec{k}_2 = \frac{1}{\sqrt{2}} [S_A^2 \quad S_B^2]^T$
- Polarimetric coherency matrices: $[T_1] = \langle \vec{k}_1 \cdot \vec{k}_1^{*T} \rangle = \begin{bmatrix} S_A^1 S_A^{1*} & S_A^1 S_B^{1*} \\ S_B^1 S_A^{1*} & S_B^1 S_B^{1*} \end{bmatrix}$, $[T_2] = \langle \vec{k}_2 \cdot \vec{k}_2^{*T} \rangle = \begin{bmatrix} S_A^2 S_A^{2*} & S_A^2 S_B^{2*} \\ S_B^2 S_A^{2*} & S_B^2 S_B^{2*} \end{bmatrix}$
- Polarimetric interferometric matrix: $[\Omega] = \langle \vec{k}_1 \cdot \vec{k}_2^{*T} \rangle = \begin{bmatrix} S_A^1 S_A^{2*} & S_A^1 S_B^{2*} \\ S_B^1 S_A^{2*} & S_B^1 S_B^{2*} \end{bmatrix}$

1. Pol-InSAR Matrices: The Dual-Pol Case

$$[T_1] = \langle \vec{k}_1 \cdot \vec{k}_1^{*T} \rangle, \quad [T_2] = \langle \vec{k}_2 \cdot \vec{k}_2^{*T} \rangle, \quad [\Omega] = \langle \vec{k}_1 \cdot \vec{k}_2^{*T} \rangle$$

2. Pre-Whitening (normalization)

- Pre-whitening (normalization):

$$[\Pi] = [T]^{-\frac{1}{2}} [\Omega] [T]^{-\frac{1}{2}}$$

- Calculate $[T] = \frac{1}{2} ([T_1] + [T_2])$, where $[T]^{\frac{1}{2}} [T]^{\frac{1}{2}} = [T]$
- Calculate the inverse $[T]^{-\frac{1}{2}}$ (see A.1)
- Whiten: $[\Pi] = \begin{bmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{bmatrix}$, with $\text{Tr}([\Pi]) = \Pi_{11} + \Pi_{22}$

- Interferometric Coherence from $[\Pi]$:

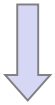
$$\tilde{\gamma}(\vec{w}_1, \vec{w}_2) = \langle \vec{w}_1^{*T} [\Pi] \vec{w}_2 \rangle$$

Geometrical Representation: The Coherence Region

- Pre-whitened Coherency Matrix: $[\Pi] = [T]^{-\frac{1}{2}}[\Omega][T]^{-\frac{1}{2}}$, with $[T] = \frac{1}{2}([T_1] + [T_2])$
- Interferometric Coherence: $\tilde{\gamma}(\vec{w}_1, \vec{w}_2) = \langle \vec{w}_1^{*T} [\Pi] \vec{w}_2 \rangle$

3. Boundary of the Coherence Region:

It can be reconstructed by estimating for each angle φ the $\max(\lambda_1)$ & $\min(\lambda_2)$ coherences.



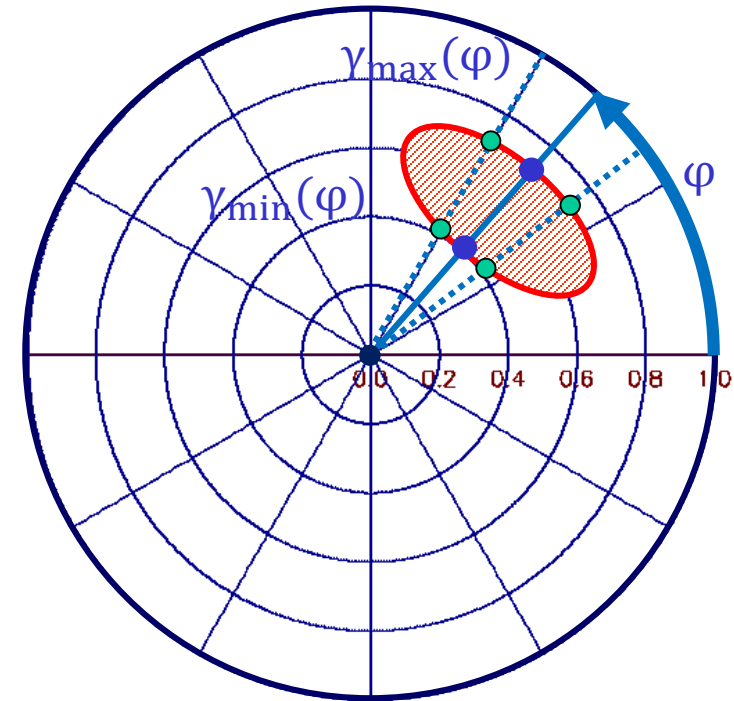
1. Optimization problem, $\vec{w}_1 = \vec{w}_2$, as an eigenvalue equation:

$$[\Pi_\varphi] \vec{w} = \lambda \vec{w}$$

$$\text{where } [\Pi_\varphi] = \frac{1}{2}([\Pi]e^{i\varphi} + [\Pi]^+e^{-i\varphi})$$

$$0 \leq \varphi \leq \pi$$

$$\lambda = -(\lambda_1 + \lambda_2^*)$$



2. Coherence Region:

Eigenvalues & Eigenvectors

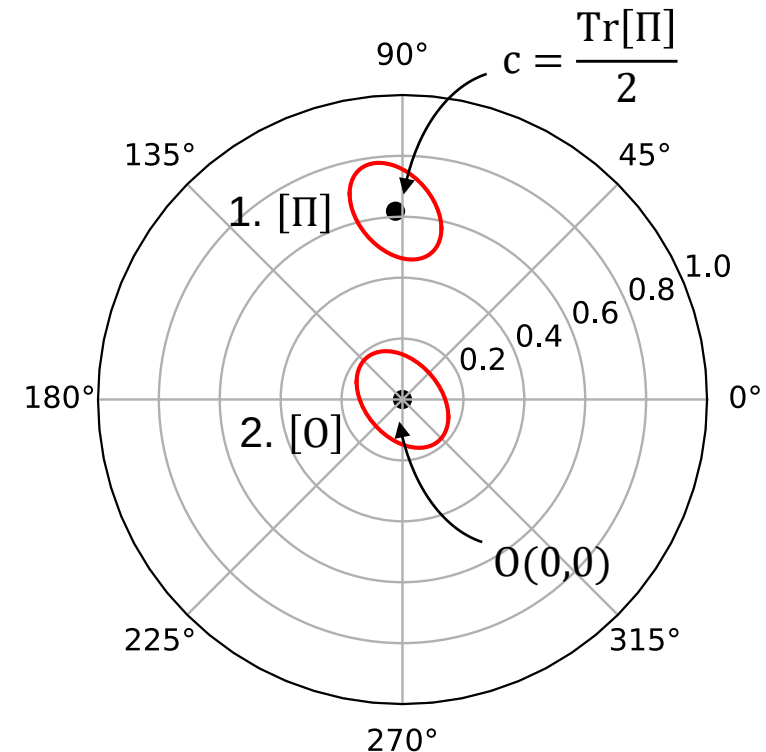
$$\forall \varphi \rightarrow \begin{cases} \lambda_{\max}, & \vec{w}_{\max} \rightarrow \gamma_{\max}(\varphi) = \vec{w}_{\max}^{*T} [\Pi] \vec{w}_{\max} \\ \lambda_{\min}, & \vec{w}_{\min} \rightarrow \gamma_{\min}(\varphi) = \vec{w}_{\min}^{*T} [\Pi] \vec{w}_{\min} \end{cases}$$

Geometrical Representation: The Coherence Region

- Pre-whitened Coherency Matrix: $[\Pi] = [T]^{-\frac{1}{2}}[\Omega][T]^{-\frac{1}{2}}$, with $[T] = \frac{1}{2}([T_1] + [T_2])$

4. Geometry of the Coherence Region:

- Center of the Coherence Region: $c = \frac{\text{Tr}[\Pi]}{2}$
- Move to the origin: $[O] = [\Pi] - c[I]$,
where $\text{Tr}([O]) = O_{11} + O_{22} = 0$
- Eigenvectors of $[O]$: $\vec{v}_1, \vec{v}_2$ (see A.2)
- Off-diagonalization of $[O]$: $r, s \in \mathbb{C}$ (see A.3)
 - Unitary matrix $[U]$ that makes the diagonal elements of $[O]$ zeros:
 - $[U]^* [O] [U] = \begin{bmatrix} 0 & r \\ s & 0 \end{bmatrix}$, $r, s \in \mathbb{C}$
 - Two complex numbers r, s (four real parameters) describe the geometry of the Coherence Region.



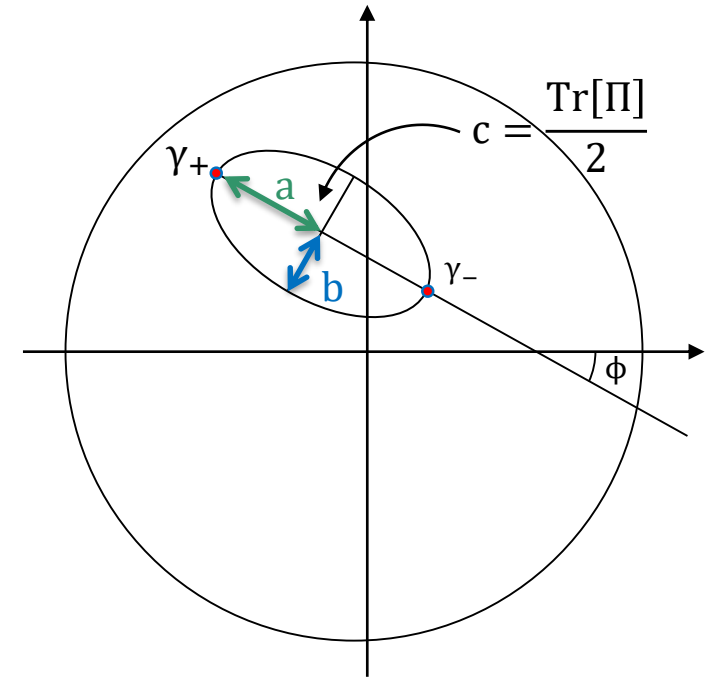
Geometrical Representation: The Coherence Region

- Pre-whitened Coherency Matrix: $[\Pi] = [T]^{-\frac{1}{2}}[\Omega][T]^{-\frac{1}{2}}$, with $[T] = \frac{1}{2}([T_1] + [T_2])$

4. Geometry of the Coherence Region:

- Center of the Coherence Region: $c = \frac{\text{Tr}[\Pi]}{2}$
- Move to the origin: $[O] = [\Pi] - c[I]$,
where $\text{Tr}([O]) = O_{11} + O_{22} = 0$
- Eigenvectors of $[O]$: $\vec{v}_1, \vec{v}_2$ (see A.2)
- Off-diagonalization of $[O]$: $r, s \in \mathbb{C}$ (see A.3)
- Obtain the Coherence Region geometry:

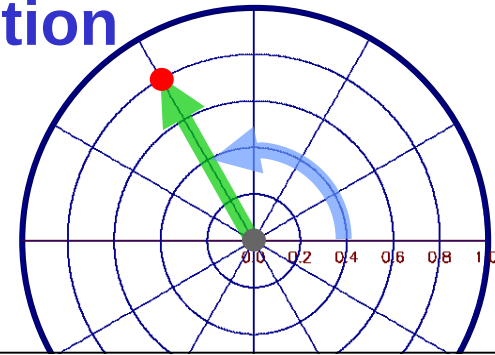
- Major axis: $2a = |r| + |s|$
- Minor axis: $2b = ||r| - |s||$
- Orientation (direction of major axis w.r.t. the x-axis): $\phi = \frac{1}{2}(\arg r + \arg s)$
- Vertices of $[\Pi]$: $\gamma_{\pm} = \frac{\text{Tr}[\Pi]}{2} \pm a e^{i\phi}$



5. Coherence Optimization: The Polar Decomposition

Interferometric Coherence: $\tilde{\gamma}(\vec{w}_i, \vec{w}_i) = \underbrace{|\tilde{\gamma}(\vec{w}_i, \vec{w}_i)|}_{\text{Radius}} \cdot \exp(i \underbrace{\text{Arg}\{\tilde{\gamma}(\vec{w}_i, \vec{w}_i)\}}_{\text{Angle}})$

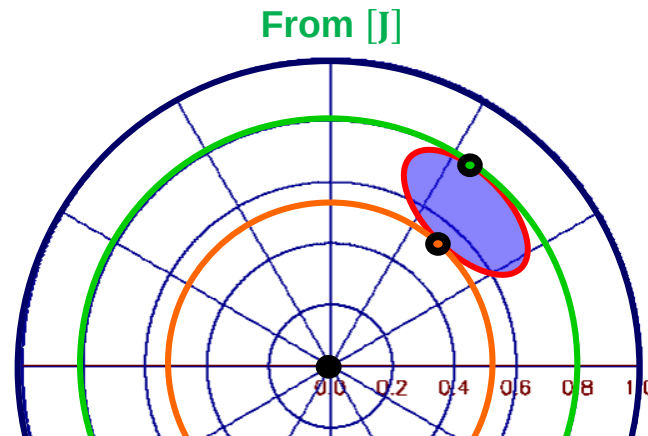
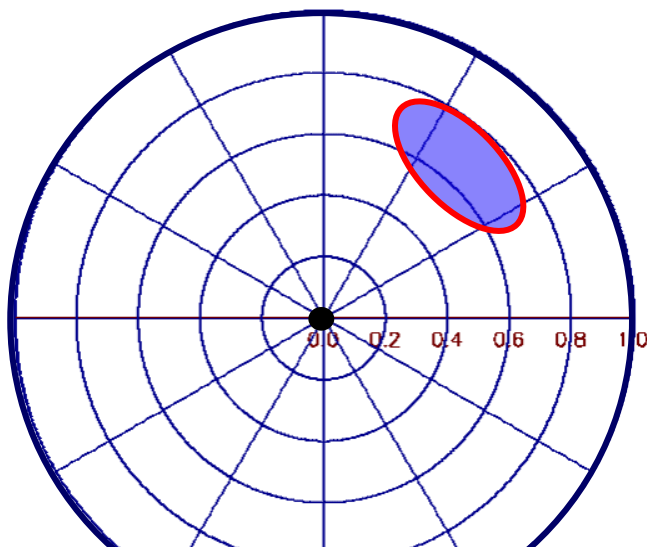
► can be represented in matrix notation with the Polar Decomposition.



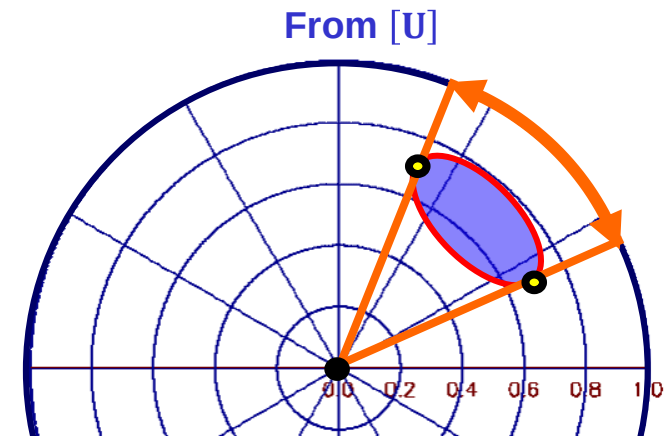
Polar Decomposition: $[\Pi] = [U] \cdot [J]$, where $\begin{cases} [J] = ([\Pi]^{*T}[\Pi])^{\frac{1}{2}} & \text{(positive semi-definite Hermitian matrix)} \\ [U] = [\Pi][J]^{-1} & \text{(unitary matrix)} \end{cases}$

► **Amplitude** can be maximized/minimized by the eigenvectors of $[J]$: $\gamma_{\max}, \gamma_{\min}$

► **Phase** can be maximized/minimized by the eigenvectors of $[U]$: $\gamma(\phi_{\min}), \gamma(\phi_{\min})$



Max./ Min. Interferometric Coherence
as function of $\vec{w}_i$



Max. Phase Difference between
interferograms formed with $\vec{w}_i$ and $\vec{w}_j$

5. Coherence Optimization: The Polar Decomposition

Polar Decomposition: $[\Pi] = [U] \cdot [J]$, where $\begin{cases} [J] = ([\Pi]^*{}^T[\Pi])^{\frac{1}{2}} & \text{(positive semi-definite Hermitian matrix)} \\ [U] = [\Pi][J]^{-1} & \text{(unitary matrix)} \end{cases}$

1. Eigenvalues and eigenvectors of the Hermitian matrix $[J]^2 = [\Pi]^*{}^T[\Pi]$ (see A.2)

- Real positive eigenvalues: $\lambda_1, \lambda_2 \in \mathbb{R}^+$
- Orthogonal eigenvectors: $\vec{v}_1 = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} \rightarrow [V] = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$

2. Optimized coherences in **amplitude**:

$$[V]^*{}^T[\Pi][V] = \begin{bmatrix} v_{11}^* & v_{21}^* \\ v_{12}^* & v_{22}^* \end{bmatrix} \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} \gamma_{\max} & \gamma_{12} \\ \gamma_{21} & \gamma_{\min} \end{bmatrix}$$

$$\gamma_{\max} = v_{11}^* v_{11} \pi_{11} + v_{21}^* v_{11} \pi_{21} + v_{11}^* v_{21} \pi_{12} + v_{21}^* v_{21} \pi_{22}$$

$$\gamma_{\min} = v_{12}^* v_{12} \pi_{11} + v_{22}^* v_{12} \pi_{21} + v_{12}^* v_{22} \pi_{11} + v_{22}^* v_{22} \pi_{22}$$

5. Coherence Optimization: The Polar Decomposition

Polar Decomposition: $[\Pi] = [U] \cdot [J]$, where $\begin{cases} [J] = ([\Pi]^{*T}[\Pi])^{\frac{1}{2}} & \text{(positive semi-definite Hermitian matrix)} \\ [U] = [\Pi][J]^{-1} & \text{(unitary matrix)} \end{cases}$

3. Eigenvalues and eigenvectors of the unitary matrix $[U] = [\Pi][J]^{-1}$ (see A.1, A.2)

- Real positive eigenvalues: $\lambda_1, \lambda_2 \in \mathbb{R}^+$
- Orthogonal eigenvectors: $\vec{w}_1 = \begin{bmatrix} w_{11} \\ w_{12} \end{bmatrix}, \vec{w}_2 = \begin{bmatrix} w_{21} \\ w_{22} \end{bmatrix} \rightarrow [W] = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$

4. Optimized coherences in **phase**:

$$[W]^{*T}[\Pi][W] = \begin{bmatrix} w_{11}^* & w_{21}^* \\ w_{12}^* & w_{22}^* \end{bmatrix} \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix} \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} \gamma(\phi_{\max}) & \gamma_{12} \\ \gamma_{21} & \gamma(\phi_{\min}) \end{bmatrix}$$

$$\gamma(\phi_{\max}) = w_{11}^* w_{11} \pi_{11} + w_{21}^* w_{11} \pi_{21} + w_{11}^* w_{21} \pi_{12} + w_{21}^* w_{21} \pi_{22}$$

$$\gamma(\phi_{\min}) = w_{12}^* w_{12} \pi_{11} + w_{22}^* w_{12} \pi_{21} + w_{12}^* w_{22} \pi_{11} + w_{22}^* w_{22} \pi_{22}$$

Appendix 1. Calculation of Inverse Matrix $[T]^{-\frac{1}{2}}$

Given a 2×2 Hermitian matrix $[T]$, $[T]^{-\frac{1}{2}}$ is computed by means of an Eigen-Decomposition:

$$[T] = \begin{bmatrix} t_{11} & t_{12} \\ t_{12}^* & t_{22} \end{bmatrix}$$

1. Eigenvalues:

$$\lambda = \sigma \pm \delta$$

$$\text{with } \sigma = \frac{\text{Tr}[T]}{2} = \frac{t_{11} + t_{22}}{2}, \quad \delta = \sqrt{\sigma^2 - D}, \quad \text{and } D = \det[T] = t_{11}t_{22} - t_{12}t_{12}^*$$

2. Eigenvectors:

$$\Rightarrow \text{For } \lambda = \sigma + \delta: \quad \vec{v}_+ = \frac{1}{\sqrt{t_{12}t_{12}^* + \left(\frac{t_{11}-t_{22}}{2} - \delta\right)^2}} \begin{bmatrix} -t_{12} \\ \frac{t_{11}-t_{22}}{2} - \delta \end{bmatrix}$$

$$\Rightarrow \text{For } \lambda = \sigma - \delta: \quad \vec{v}_- = \frac{1}{\sqrt{t_{12}t_{12}^* + \left(\frac{t_{11}-t_{22}}{2} - \delta\right)^2}} \begin{bmatrix} \frac{t_{11}-t_{22}}{2} - \delta \\ t_{12}^* \end{bmatrix}$$

Appendix 1. Calculation of Inverse Matrix $[T]^{-\frac{1}{2}}$

$$[T] = \begin{bmatrix} t_{11} & t_{12} \\ t_{12}^* & t_{22} \end{bmatrix} = [\vec{v}_+ \quad \vec{v}_-] \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix} [\vec{v}_+ \quad \vec{v}_-]^*{}^T$$

3. Inverse $[T]^{-\frac{1}{2}}$:

$$[T]^{-\frac{1}{2}} = [\vec{v}_+ \quad \vec{v}_-] \begin{bmatrix} \frac{1}{\sqrt{\lambda_+}} & 0 \\ 0 & \frac{1}{\sqrt{\lambda_-}} \end{bmatrix} [\vec{v}_+ \quad \vec{v}_-]^*{}^T$$

$$[T]^{-\frac{1}{2}} = \frac{1}{t_{12}t_{12}^* + \left(\frac{t_{11} - t_{22}}{2} - \delta\right)^2} \begin{bmatrix} \frac{t_{12}t_{12}^*}{\sqrt{\lambda_+}} + \frac{\left(\frac{t_{11} - t_{22}}{2} - \delta\right)^2}{\sqrt{\lambda_-}} & t_{12} \left(\frac{1}{\sqrt{\lambda_-}} - \frac{1}{\sqrt{\lambda_+}} \right) \left(\frac{t_{11} - t_{22}}{2} - \delta \right) \\ t_{12}^* \left(\frac{1}{\sqrt{\lambda_-}} - \frac{1}{\sqrt{\lambda_+}} \right) \left(\frac{t_{11} - t_{22}}{2} - \delta \right) & \frac{t_{12}t_{12}^*}{\sqrt{\lambda_-}} + \frac{\left(\frac{t_{11} - t_{22}}{2} - \delta\right)^2}{\sqrt{\lambda_+}} \end{bmatrix}$$

Appendix 2. Eigenproblem Solution of Matrix [A]

Given a generic complex-valued 2×2 matrix [A]: $[A] = \begin{bmatrix} a_{11} & a_{12} \\ a_{12}^* & a_{22} \end{bmatrix}$,

the Eigenproblem equation is: $[A]\vec{v} = \lambda\vec{v}$

1. Eigenvalues:

$$\lambda_{\pm} = \frac{1}{2} \left((a_{11} + a_{22}) \pm \sqrt{(a_{11} - a_{22})^2 + 4a_{12}a_{21}} \right)$$
$$\lambda_{\pm} = \pm \sqrt{a_{11}^2 + a_{12}a_{21}} = \pm \alpha$$

2. Eigenvectors:

➤ For $\lambda = +\alpha$: $\vec{v}_+ = \frac{1}{\|\vec{\alpha}_+\|} \vec{\alpha}_+ = \begin{bmatrix} v_{+1} \\ v_{+2} \end{bmatrix}$ with $\vec{\alpha}_+ = \begin{bmatrix} -a_{12} \\ a_{11} - \sqrt{a_{11}^2 + a_{12}a_{21}} \end{bmatrix}$

➤ For $\lambda = -\alpha$: $\vec{v}_- = \frac{1}{\|\vec{\alpha}_-\|} \vec{\alpha}_- = \begin{bmatrix} v_{-1} \\ v_{-2} \end{bmatrix}$ with $\vec{\alpha}_- = \begin{bmatrix} a_{22} + \sqrt{a_{11}^2 + a_{12}a_{21}} \\ -a_{21} \end{bmatrix}$

Appendix 3. Off-diagonalization of Matrix [O]

- Goal: Find a Unitary Matrix [U] that makes the diagonal elements of $[O] = \begin{bmatrix} o_{11} & o_{12} \\ o_{12}^* & o_{22} \end{bmatrix}$ zeros.
- Lemma: Unitary transform does not change the coherence region.

1. Eigenvectors of [O]: $\vec{v}_1, \vec{v}_2$ (see A.2)

$$\vec{v}_+ = \frac{1}{\|\vec{\alpha}_+\|} \vec{\alpha}_+ = \begin{bmatrix} v_{+1} \\ v_{+2} \end{bmatrix} \quad \text{with} \quad \vec{\alpha}_+ = \begin{bmatrix} -o_{12} \\ o_{11} - \sqrt{o_{11}^2 + o_{12}o_{21}} \end{bmatrix}$$
$$\vec{v}_- = \frac{1}{\|\vec{\alpha}_-\|} \vec{\alpha}_- = \begin{bmatrix} v_{-1} \\ v_{-2} \end{bmatrix} \quad \text{with} \quad \vec{\alpha}_- = \begin{bmatrix} o_{22} + \sqrt{o_{11}^2 + o_{12}o_{21}} \\ -o_{21} \end{bmatrix}$$

2. Assume there is an angle θ that satisfies:

$$(e^{i\theta}\vec{v}_+ + \vec{v}_-)^{*T} [O] (e^{i\theta}\vec{v}_+ + \vec{v}_-) = 0 \rightarrow \begin{bmatrix} e^{i\theta}v_{+1} + v_{-1} \\ e^{i\theta}v_{+2} + v_{-2} \end{bmatrix}^{*T} \begin{bmatrix} o_{11} & o_{12} \\ o_{12}^* & o_{22} \end{bmatrix} \begin{bmatrix} e^{i\theta}v_{+1} + v_{-1} \\ e^{i\theta}v_{+2} + v_{-2} \end{bmatrix} = 0$$

Appendix 3. Off-diagonalization of Matrix [O]

2. Assume there is an angle θ that satisfies:

$$(e^{i\theta}\vec{v}_+ + \vec{v}_-)^{*T} [O] (e^{i\theta}\vec{v}_+ + \vec{v}_-) = 0 \rightarrow \begin{bmatrix} e^{i\theta}v_{+1} + v_{-1} \\ e^{i\theta}v_{+2} + v_{-2} \end{bmatrix}^{*T} \begin{bmatrix} o_{11} & o_{12} \\ o_{12}^* & o_{22} \end{bmatrix} \begin{bmatrix} e^{i\theta}v_{+1} + v_{-1} \\ e^{i\theta}v_{+2} + v_{-2} \end{bmatrix} = 0$$

3. Solution: $2\alpha i \text{Im}\{e^{-i\theta} \cdot \vec{v}_+^{*T} \cdot \vec{v}_-\} = 0 \rightarrow 2\alpha i \text{Im}\left\{e^{-i\theta} \begin{bmatrix} v_{+1} \\ v_{+2} \end{bmatrix}^{*T} \begin{bmatrix} v_{-1} \\ v_{-2} \end{bmatrix}\right\} = 0$

$$\theta = \arg(v_{+1}^* v_{-1} + v_{+2}^* v_{-2}) = \text{angle}(\vec{v}_+^{*T} \cdot \vec{v}_-)$$

4. The unitary matrix [U] that makes the diagonal elements of [O] zeros is now given by

$$[U] = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}$$

$$\text{with } \begin{bmatrix} u_{11} \\ u_{21} \end{bmatrix} = \frac{1}{\|e^{i\theta}\vec{v}_+ + \vec{v}_-\|} \begin{bmatrix} e^{i\theta}v_{+1} + v_{-1} \\ e^{i\theta}v_{+2} + v_{-2} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} u_{12} \\ u_{22} \end{bmatrix} = \frac{1}{\|e^{i\theta}\vec{v}_+ - \vec{v}_-\|} \begin{bmatrix} e^{i\theta}v_{+1} - v_{-1} \\ e^{i\theta}v_{+2} - v_{-2} \end{bmatrix}$$

Appendix 3. Off-diagonalization of Matrix $[O]$

5. The off-diagonal form of $[O]$ is now $[U]^*{}^T[O][U] = \begin{bmatrix} 0 & r \\ s & 0 \end{bmatrix}$, where $r, s \in \mathbb{C}$:

$$[U]^*{}^T[O][U] = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}^{*T} \begin{bmatrix} o_{11} & o_{12} \\ o_{21} & o_{22} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} 0 & r \\ s & 0 \end{bmatrix}$$

$$\begin{cases} r = o_{12} = o_{11}(u_{11}^* u_{12} - u_{21}^* u_{22}) + o_{12} u_{11}^* u_{22} + o_{21} u_{21}^* u_{12} \\ s = o_{21} = o_{11}(u_{12}^* u_{11} - u_{22}^* u_{21}) + o_{12} u_{12}^* u_{21} + o_{21} u_{22}^* u_{11} \end{cases}$$

The Pol-InSAR Course

Part 7: Pol-InSAR 1

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